Program overview

14-Dec-2022 12:21

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Master Nanobiology**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master NB 2022	**Master Nanobiology 2022**		
1st year Msc NB 2022			
AP3162	Physics of Biological Systems: Mathematical Modelling in Systems Biology	6	
NB4011	Analytical Mechanics	3	
NB4012	Stochastic Processes with Applications	3	
NB4020	High-Resolution Imaging	4	
NB4030	Engineering Genetic Information	3	
NB4040	Biology of Cancer	4	
NB4050	Modeling Dynamical Systems	3	
NB4070	Soft Matter	6	
NB4510	Proposal Development Part 1	3	
NB4520	Proposal Development Part 2	7	
NB5015	Seminars	2	
Electives Nanobiology			
NB4080	Protein Quality Control Mechanisms	3	
NB4090	Stem Cells	3	
NB4100	Nuclear Architecture	3	
NB4110	Geometry of Physics	6	
NB4120	Biological Networks; a data driven approach to discovery and understanding	3	
NB4150	The Origin and Synthesis of Life	6	
NB4160	Engineering of living systems	3	
NB4165	Molecular Virology & Immunology	3	
NB5910	Independent Research	0	
Other Electives Courses 2022			
AP3021	Advanced Statistical Mechanics	6	
AP3032	Continuum Physics	6	
AP3132	Advanced Digital Image Processing	6	
AP3162	Physics of Biological Systems: Mathematical Modelling in Systems Biology	6	
AP3232	Medical Imaging Signals and Systems	6	
AP3371	Radiological Health Physics	6	
AP3582	Medical Physics of Photon and Proton Therapy	6	
AP3831	Systems Engineering for Physicists	3	
BM41035	Biomaterials	4	
BM41050	Applied Experimental Methods: Medical Instruments	4	
BM41075	Regenerative Medicine	4	
BM41090	Computational Mechanics of Tissues and Cells	6	
BM41155	3D Printing	4	
CS4220	Machine Learning 1	5	
CS4230	Machine Learning 2	5	
CS4255	Algorithms for sequence-based Bioinformatics	5	
CS4329	Recent topics in bioinformatics	5	
EE4650	Advanced Magnetic Resonance Imaging	5	
IFEEMCS4250	Statistical Learning for Engineers	4	
IN4089	Data Visualization	5	
LM3311	Green Chemistry and Sustainable Technology	3	
LM3432	Analysis of Metabolic Networks	6	
LM3442	Metabolic Reprogramming	6	
LM3451	Bioprocess Integration	5	
LM3561	Ethical, Legal and Social Issues in Biotechnology	3	
LM3581NB	Metabolic Systems Biology	3	
LM3601	Molecular Biotechnology & Genomics	6	
LM3611	Microbial Community Engineering	6	
LM3701	Advanced Enzymology	6	
LM3741	Fermentation Technology & Environmental Biotechnology	6	
LM3751	Transport & Separation	6	
LM3771	Protein Engineering	3	
ME45025	Introduction to Multiphase Flow	5	
ME45043	Advanced Fluid Dynamics for AP	6	
ME46000	Nonlinear Mechanics	4	
ME46072	Nonlinear Dynamics	4	
		3	

SC42030	Control for High Resolution Imaging	
TPM305A	Writing a Masters Thesis in English	2
WI4011-17	Computational Fluid Dynamics	6
WI4014TU	Numerical Analysis	6
WI4019-SP	Non-linear Differential Equations	6
WI4201	Scientific Computing	6
WI4204	Advanced Modeling	6
WI4205	Applied Finite Elements	6
WI4212	Advanced Numerical Methods	6
WI4430	Martingales, Brownian Motion	6
WM-ITAV-4010	Scientific Writing	2
WM0201TU-ENG	Technical Writing	2
WM0320TU	Ethics and Engineering	3

2nd year Msc NB 2022

NB5015	Seminars	2
NB5020	Literature Review Report	4
NB5030	Project Proposal Writing	2
NB5040	Research Presentation	2
NB5900	Master End Project Nanobiology	44
NB5942	Master End Project Nanobiology	42

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Master Nanobiology**

Master NB 2022

Responsible Program Employee	J.C. Colgrove
Director of Studies	T. Idema
Program Coordinator	D. Mevius
Program Coordinator	J.C. Colgrove
Contact for students	Info-MSc-NB@tudelft.nl
In association with the University of	Erasmus University Rotterdam, Faculty of Medicine (Erasmus MC)
Prerequisites	Applicants are required to have a similar background to what the Nanobiology Bachelor programme provides. Bridging programmes may be available for applicants with a relevant Dutch university background or who are residents of the Netherlands with a non-Dutch university background.
Introduction 1	In Nanobiology, we integrate methods and theories from physics, and molecular biology, using tools from math and computer modelling to further understanding of the molecular basis of life. What youll learn in the Nanobiology master has direct applications in research. Developments in biomedicine, such as studies on human genome variation and the control of stem cells, increasingly require analysis and quantitative description at the fundamental level. Moreover, it is becoming possible to develop artificial biomolecules and nanoparticles with wide applications in research and medicine. The incorporation of new biological building blocks is highly promising in for instance industrial biotechnology and medical science. These advances will reshape many aspects of medical diagnosis and treatment. The rapid advancement of modern biomedical, biophysical and computational technologies promises to provide new tools to gain in depth knowledge of the fundamental molecular and cellular mechanisms controlling health or involved in disease. The programme provides a strong foundation, and includes a great deal of flexibility with electives and research projects so students can customize to suit their particular research interests.
Program Goals	The objective of the Masters degree programme [Nanobiology] is to providing students with the knowledge, insight and skills that enables them to perform independent professional and scientific activities in the area of Nanobiology at an academic level. 5.2.a. Knowledge 1 The student has theoretical and practical knowledge of the physics of biological processes and the methods to observe them. 2 The student is able to build mathematical models of physical and biological systems, and can solve them numerically and/or analytically. 3 The student can apply their knowledge to quantify biological processes from experimental results. 4 The student understands ethical issues in research. 5.2.b Research skills 5 The student is able to identify a problem and translate this into a research question. 6 The student is able to conduct comprehensive literature investigations, related to the research question. 7 The student is able to translate a research question into a research proposal. 8 In collaboration with other research group members, the student is able to set up and conduct a research project, collect data, analyse data, and come to conclusions. 9 The student is able to perform ethically responsible research. 5.2.c Communication skills 10 The student is able to write research findings in the form of a draft manuscript, which in collaboration with a research supervisor may be developed into a scientific article, suitable for publication in an international, peer-reviewed journal. 11 The student can communicate his or her results in oral and written form to audiences of specialists and non-specialists. 12. The student understands ethical considerations in communicating science.
Program Structure 1	The programme has a study load of 120 credits. It consists of 36 EC of mandatory coursework including a choice between two Biophysics courses. 22 EC of elective courses. 18 EC broadening research project (choose from 5 options). One option requires an additional 3 EC of electives. 42 EC Master End Project (MEP). This large project serves to prove that the student has a good command of, and is able to apply, all the knowledge, insight and skills gained during the Master's degree programme. (Note this changed beginning in 2022-2023, previous cohorts had a slightly different distribution of credits).
Administration by the Faculty of	TNW
Gives access to	Students who have successfully completed their Masters degree programme and have consequently been granted the title Master of Science in Nanobiology are eligible to follow PhD research programmes and are qualified to fulfil positions in a plethora of labour market sectors: all science related segments of the labour market, including research, development, and education. Because of their thorough training in general problem solving they can easily adapt themselves to challenges in fields outside the strict science domain and therefore they are quickly employable in non-science related jobs.

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1st year Msc NB 2022	
Responsible Program Employee	J.C. Colgrove
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Program Coordinator	D. Mevius
Program Coordinator	J.C. Colgrove
Introduction 1	The list below is of obligatory courses. It includes a pair of courses which students make a choice between. Students must take either: AP3162 Physics of Biological Systems: Mathematical Modeling in Systems Biology or NB4070 Soft Matter Students can decide to take both, and then AP3162 Phyisics of Biological Systems will be counted as an elective.

AP3162	Physics of Biological Systems: Mathematical Modelling in Systems Biology	6
Responsible Instructor	Dr.ir. L. Laan	
Instructor	Dr.ir. S.J. Tans	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic differential equations. Not necessary to have any prior knowledge of biology.	
Course Contents	This is a course intended for physicists, biologists, engineers, and quantitative-minded students who are fascinated by the questions: "What is life?" and "What are the physical and mathematical rules that govern living cells and tissues?" This course takes a quantitative approach to understand genetic circuits, pattern formation, and molecular recogniton systems. If you are a physicist and have wondered why so many physicists are entering biology, this course will hopefully answer your question in depth. If you are a biologist and have wanted to study biology with mathematical models and quantitative data, then this course will hopefully teach you how to do so. In recent years there has been a tremendous progress in revealing the fascinating internal dynamics of cells. The driving promise is to understand living systems bottom-up, by combining existing knowledge of individual proteins and reactions, novel techniques to follow single cells and molecules in time, and simple mathematical models to explain the organisation and dynamics. In this course we will focus on the key innovative concepts that have been discovered using model systems. The course aims to cover the following topics: basic biology, pattern formation inside cells, gene regulation, temporal and spatial dynamics, stochasticity and noise, evolution, genotype to function relations and cytoskeletal dynamics Students will also explore results of cutting edge, modern research, where novel experimental or theoretical approaches are used to investigate central questions in biology. By reading and discussing research papers that are just months or 1-2 years old, the students will get to directly apply their knowledge from this course to research papers in this fast moving field driven by engineers, physicists, and quantitative biologists.	
Study Goals	- To be knowledgeable about how math and physics can be applied to study cellular processes. - To be knowledgeable about modelling the following key concepts:, pattern formation, stochasticity and noise in cells, evolutionary dynamics, temporal and spatial organization of cellular processes, cytoskeletal dynamics. - To be able to design and interpret a quantitative experiment to address concepts listed above. - To be knowledgeable about current research literature in quantitative biology and physics of life.	
Education Method	Lectures and one-on-one discussions with the instructors and TAs Students presentations	
Literature and Study Materials	We will provide the study material in class.	
Assessment	The final grade for the course will be determined as follows: - report about a paper (30%) - presentation about a paper (30%) - Final exam (40%) In order to pass this course, students must obtain at least a passing grade (5,8) on the final exam and an overall grade of at least 5,8.	
Permitted Materials during Tests	all material handed out to you during the course	
Elective	Yes	

NB4011	Analytical Mechanics	3
Responsible Instructor	Dr. J.L.A. Dubbeldam	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course deals with Lagrange's and Hamilton's formalism of classical mechanics. Subjects discussed are: - From Newtons law to Lagrangian mechanics and least action principle. Euler-Lagrange equations, generalized momenta. - Constrained systems, Lagrange multipliers. - Generalized coordinates, rigid body, inertia tensor, harmonic oscillators, Poisson brackets. Symmetries of the Lagrangian and conserved quantities. - Hamiltons principle, Hamiltonian, Hamiltons equations, Liouville's Theorem.	
Study Goals	1. Understanding and applying Lagrange formalism in a concrete context. Being able to derive the Euler Lagrange equations for a given Lagrangian and understand their meaning in various (biology relevant) contexts. Understand the meaning of the different terms constituting the Lagrangian. 2. Understanding the link between Hamilton and Lagrangian formalism. Understanding the meaning of Hamiltonian, being able to derive Hamiltons equations in various contexts. Understanding the role of symmetries, and being able to apply the connection with conserved quantities.	
Education Method	Lectures and workgroups	
Literature and Study Materials	Lecture notes will be provided during the course. Also the lecture of David Tong, Classical Mechanics, will be used.	
Assessment	Homework assignments and written exam. The homework grade is determined by the best 2 homeworks, out of 3. The final grade is calculated as max(0.75*exam+0.25*homework,exam). The homework grades also count for the resit, but cannot be transferred to next year.	
Location	TU Delft	

NB4012	Stochastic Processes with Applications	3
Responsible Instructor	Dr. E. Pulvirenti	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2A 2B 3B	
Course Language	English	
Course Contents	This course deals with Markov processes. Subjects are: - Discrete and continuous-time Markov processes on finite and discrete state spaces: transition probabilities, master equation, stationary distribution, ergodicity, random walks, Markov chain Monte Carlo method. - Diffusion processes: Brownian motion, diffusion equation, diffusion with drift, reversible diffusions, stationary distribution, Fokker-Planck equation.	
Study Goals	By the end of the course, you should be able to: 1. Explain the theory of Markov processes in discrete and continuous time on finite and discrete state spaces. 2. Analyse a Markov process and recognize whether it has a unique/non-unique stationary distribution and whether it is ergodic. 3. Explain the theory of processes with continuous state space such as jump and diffusion processes. 4. Investigate how to model a system via a Markov process or a master equation. 5. Investigate how to model a system with a diffusion process and apply the connection between diffusion processes and partial differential equations of diffusion type.	
Education Method	Lectures and workgroups	
Assessment	Homework assignments and written exam. The homework grade is determined by the best 2 homeworks, out of 3. The final grade is calculated as max(0.8*exam+0.2*homework,exam). The homework grades also count for the resit, but cannot be transferred to next year.	
Location	TU Delft	

NB4020	High-Resolution Imaging	4
Responsible Instructor	Dr. A. Jakobi	
Instructor	Dr. B. Rieger	
Instructor	Dr.ir. J.P. Hoogenboom	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Microscopy is an essential technique in the life science and developments in microscopy have almost always led to new biological discoveries. In the course High Resolution Imaging we introduce the state of the art techniques to observe cells, tissue, and biological molecules at the highest possible resolution. We provide a working knowledge of various high resolution imaging techniques, present the basic principles and limitations behind these techniques, and also discuss how to prepare biological samples for the different forms of microscopy. Examples from recent scientific literature will be used to illustrate what can be and has been achieved. Targeted homework, lab demonstrations, and experimental as well as theoretical exercises will help to familiarize you with the different types of microscopes and the data they provide. Specifically, the course includes lectures and demos on: Superresolution fluorescence microscopy. Scanning electron microscopy (SEM), Transmission electron microscopy (TEM), and Scanning transmission electron microscopy (STEM). Electron cryo-microscopy, and methods to calculate 3D structures from 2D projections. Correlative light and electron microscopy and techniques for large-scale and volume imaging.	
Study Goals	After completion of this course, the students should: 1. Understand the basic principles and limitations for focusing light and electron beams in microscopes. 2. Be able to explain the principles for contrast and image formation in the SEM, STEM, and TEM. 3. Understand the properties of fluorescent labels and the basic principles used in superresolution light microscopy 4. Understand the basic principles of 3D electron microscopy. 5. Be able to explain the role of sample preparation in high-resolution microscopy, understand the steps of fixation (chemical and cryogenic), staining, embedding, sectioning, and labelling, and reproduce the main considerations for each step. 6. Be able to estimate the limitations of each method and decide which of them would be suitable in a particular experiment. 7. Be able to judge the validity of a microscopy approach presented in an experimental work.	
Education Method	Lectures, homework, image processing exercises. If circumstances permit discussion groups, practicals and demonstrations may also be part of the course. Exact list of course activities may change due to Covid-19 regulations.	
Literature and Study Materials	Required: Lecture notes & articles provided during the course.	
Assessment	The final grade is comprised of the grades for the assignments and the exam grade where each individual grade has to be 5.5 or higher. The maximum weight for all assignments together is 30%, but the percentage may be lower if not all activities can be included because of Covid-19 restrictions. The grades for the assignments remain valid for the retake. The retake may be conducted as an oral exam if <5 students subscribe for the retake. There is no possibility to retake an assignment. Assignment grades expire at the end of the academic year. A student retaking the course in the next academic year has to do the assignments of the new course.	
Location	Delft	

NB4030	Engineering Genetic Information	3
Responsible Instructor	Dr.ir. J.H.G. Lebbink	
Instructor	Dr. J.A.A. Demmers	
Instructor	Dr. O. Zelenskyy	
Instructor	Dr. J. Pothof	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	NB1022 Genetics; NB1102 & NB1110 Chemistry 1 and 2; NB1012 Biochemistry, NB1016 Molecular Biology.	
Course Contents	This advanced course focuses on the information embedded in the genome of a cell: how it is organized, accessed, interpreted, manipulated and analyzed using structural biological, genomics and proteomics approaches. The course will introduce the students to the methods for precise modification of eukaryotic and prokaryotic genomes (genomic engineering), show how they can be used to create models for fundamental research, correct genetic defects by gene therapy, and develop new strains and varieties for agriculture and biotechnology. Strengths and limitations of both state of the art (CRISPR/Cas) and historical techniques will be described. To illustrate the required underlying principles for making informed choices during the design of a genome engineering approach, the course addresses how the structure of biomolecules defines their function, how to interpret structural information that is available in public databases, and how to use this information in the design of variants for functional studies and in mapping and interpretation of patient mutations. Furthermore, the students will learn how to analyze information encoded in the genome using genomics, transcriptomics and proteomic tools. For the latter, students will be introduced into the principles of mass spectrometry-based proteomics and will learn how to identify (mutant) proteins based on the interpretation of mass spectra in a practical course. At the end of this course, students will be able to tackle research questions using genomics, transcriptomics and proteomics concepts in an integrated approach. The focus of this course is on application and integration of knowledge. Active participation in class assignments and discussion sessions is required.	
Study Goals	At the completion of this course, the students will be able to: - Apply basic biochemical and biophysical principles to explain how protein structure determines function - Design structure-based mutations to explore biological function of proteins - Access and interpret genomic sequence and annotation using genomic browsers - Conceptually plan a modification of a mouse gene using state of the art methods - Interpret genetic models produced by commonly used genomic engineering techniques - Explain the principles and methods of genomics and transcriptomics - Design a typical genomics and/or transcriptomics experiment - Evaluate data from a typical genomics/transcriptomics experiment - Explain the basics of biomolecular mass spectrometry - Describe how mass spectrometry is used in biomedical research and biotechnology - Analyze data from a typical quantitative proteomics experiment - Design a typical (quantitative) proteomics experiment - Formulate gene function by integrating information from structural biology, proteomics, genomics, transcriptomics and genome engineering principles and techniques.	
Education Method	This course will be taught using lectures, group discussion, computer practicals and assignments. During these activities, professional skills that will also be developed are reading scientific literature, integrating and applying scientific knowledge, problem solving, and time management. The course is interactive with in-class contact hours. If necessary, online or a combination of in-class and on-line lectures will be organized depending on COVID-19 restrictions.	
Computer Use	Yes bring your laptop	
Literature and Study Materials	Slides, scientific articles, web-based tools and additional study material will be made available.	
Assessment	Computer exam on campus 100%. Format may change due to COVID-19 restrictions.	
Special Information	see Brightspace for teacher emails.	
Schedule	see MyTimeTable and Brightspace	
Location	Rotterdam and/or online	

NB4040	Biology of Cancer	4
Responsible Instructor	Prof.dr. R. Fodde	
Instructor	Prof.dr.ir. G. Jenster	
Instructor	Prof.dr. H.R. Delwel	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Prior to the beginning of the NB4040 course, students are expected to have a solid basic knowledge of the molecular and cellular biology of the cell as described in the 6th edition of the Alberts' textbook (already in your possession).	
Course Contents	Understanding cancer means understanding the multifaceted nature of biology. The reason for this rather blunt statement is straightforward: among the broad spectrum of diseases, cancer represents a unique didactic model as the malignant cell exploits to its own advantage the most essential biological features of the normal cell in development and adulthood. Thus, by studying the cancer cell in comparison to its normal equivalent, scientists have been able throughout the years to elucidate the molecular and cellular basis of life. That is why the study of the molecular and cellular basis of cancer is (or should be) integral part of any life science academic curriculum. The NB4040 course satisfies this need within the MSc program Nanobiology. The general idea behind the NB4040 course is to provide the students with a basic knowledge of the biology of cancer by taking advantage of the lead provided by the seminal paper The Hallmarks of Cancer by D. Hanahan and R. Weinberg where the most essential features of the cancer cell have been described and discussed. The paper has already been updated in 2011 and most recently in 2022, which highlights its relevance in the cancer scientific community. The book The Biology of Cancer, 2nd edition by R. A. Weinberg (Garland Science, Taylor & Francis Group, LLC, London, 2007, ISBN-10 0-8153-4078-8/ ISBN-13: 9780815340782) will form the basic textbook for the NB4040 course. The course will start with three sessions on the cellular and molecular biology of cancer and its hallmarks. This will lay the basis for the following sessions where the focus will be on specific cancer types and their features, each covered by expert scientists and clinicians from the Erasmus MC Cancer Institute. These will include leukemia, prostate, bladder, and colon cancer. An additional session will be devoted to omics technologies and bioinformatics as essential experimental tools in cancer research. These papers and book, together with any other reading material and the pdf's of the lectures are available for download from the Brightspace page of the course. The format of the NB4040 is meant to satisfy different needs of the Nanobiology MSc student: 1. For those passionate about nanotechnology and its more translational and clinical applications (think of liposomes, DNA origami, nanobots, etc.), the course will highlight how each cancer type, depending on its specific hallmarks and peculiarities, and on the anatomical location of the primary lesion and of its metastases, will need specific strategies towards the development of effective targeted approaches. 2. For those who are more interested in cellular and molecular biology in homeostasis and cancer, the course will provide ample opportunities to deepen your knowledge and appreciate how even the relative relevance of each hallmark and the underlying mechanisms can substantially differ across distinct cancer forms. In both cases, the elucidation of the hallmarks through which a cancer cell is established, progresses through different malignancy stages, and eventually metastasize distant organ sites, allows us to study and understand biology. As mentioned above, the cancer cell hijacks normal (developmental and homeostatic) programs to exploit them to its own advantage. Paradoxically, although as of today many obstacles have to be overcome to translate our current knowledge into therapies, we have nevertheless been acquiring a substantial body of fundamental knowledge in cellular and molecular biology through the study of cancer.	
Study Goals	After completion of this course, students should have acquired a multi-disciplinary and functional knowledge of the disease cancer so to be able to address both fundamental and more translational issues (e.g. cellular and molecular mechanisms underlying fundamental hallmarks; the development of nanotechnologies towards the diagnosis, prognosis, and therapy of specific tumor types). Apart from the textbook knowledge, the students will be able to 1. write a scientific report on a recently published cancer-related paper (based on the News&Views format found in Nature and all major scientific journals). 2. Understand scientific literature on recent advances in cancer biology. 3. Develop a research question based on literature research. 5. Collaborate with fellow students to prepare a presentation (and present it). 6. Critically evaluate other students' scientific presentations.	
Education Method	This course will include the following educational activities: - conventional lectures (hoorcolleges); - interactive lectures (werkcolleges); - homework (non graded) - group work: small groups of students (2-5) will be challenged by specific scientific questions, the answer to which will require knowledge of the hallmarks of cancer, active literature searches, and the preparation of the group presentation followed by Q&A sessions. As the roles will be inverted, students will also be asked to critically challenge the presentations by their peers.	
Literature and Study Materials	The Biology of Cancer, 2nd edition by R. A. Weinberg (Garland Science, Taylor & Francis Group, LLC, London, 2007, ISBN-10 0-8153-4078-8/ ISBN-13: 9780815340782). Additional original papers and reviews from the recent scientific literature will be provided through Brightspace.	
Assessment	NB4040 is a graded course. The evaluation (exam) will consists of: 1. a written open question exam (value: 50%) to assess the students general knowledge on the molecular and cellular basis of cancer (i.e. its hallmarks); and 2. a written essay of the "News & Views" type (value: 50%) on recent scientific publication (cancer- and/or hallmark-related) of choice. The written essay will test the students capacity to address and critically discuss specific issues (e.g. more fundamental questions relative to specific hallmarks; or more translational issues where nanotechnology is applied to cancer prevention,	

	prognosis, diagnosis and therapy). The written exam must be passed with a 5.8 or higher for the review to be evaluated. The re-take exam will consists of exactly the same two items as above. An oral exam may replace the written 'open questions', especially when a limited number of students enroll fo the re-sit.
Remarks	I do NOT use any tu.delft email address. I am ONLY and EXCLUSIVELY to be contacted at r.fodde@erasmusmc.nl
Location	Onderwijscentrum, Erasmus MC Rotterdam

NB4050	Modeling Dynamical Systems	3
Responsible Instructor	Ir. M.J.F. Negrello	
Contact Hours / Week x/x/x/x	4/0/0/0/	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	The course assumes knowledge in python (e.g., NB2181), calculus and linear algebra. The course benefits from insights gained from evo-devo and is complementary to 'mathematics for nanobiologists'.	
Course Contents	The course intends to train the students in the art of computational modeling, by implementing dynamical systems models. The students will be introduced to modeling through various examples and practical sessions, outlining design principles and computational requirements. Students will learn to analyze models with the concepts and tools of dynamical systems theory, such as attractors, bifurcations and parameter space analysis. The course starts with the analysis of one-dimensional dynamical systems (such as the logistic map), moving on to multiple dimensions (Lotka-Volterra and neuronal network models) and finally to spatial models, such as coupled oscillators. At the end of the course students will use 3 sessions to implement and analyze their own model of choice. They then present this model and write a report about it.	
Study Goals	At the completion of this course, the students will be able to: 1. Describe a natural phenomenon in terms of state variables, parameters and transitions, by abstracting the relevant components of a system / phenomenon. 2. Implement the various models (in Python) from first principles. 3. Conduct parameter space analysis on dynamical behai their implemented models. 4. Identify and recognize relevant dynamical phenomena such as fixed-points, saddle nodes, periodic and chaotic attractors. 5. Characterize the stability of attractor states. 6. Find the bifurcations of simple dynamical system numerically. 7. Determine the sensitivity to initial conditions for different dynamical systems.	
Education Method	The course centers around a combination of lectures, computational projects (python), and a final project. Student will gain knowledge/experience in the academic skill(s): - Scientific Writing (Report) - Presenting Scientific Theory and Results (Presentation) - Team work (final group project) - Coding in python (Assignments and Final project) - Giving useful feedback (Peer-Review of Report)	
Literature and Study Materials	Required: - Students are not required to buy literature, as lecture notes provided at the course in the form of jupyter notebooks. However, students are encouraged to obtain access to digital versions of the first two references below. Recommended (in order of relevance): 1 James Yorke, Elizabeth Alligood - Chaos (https://www.springer.com/gp/book/9780387946771) 2 Steven Strogatz - Nonlinear Dynamics and Chaos (https://www.routledge.com/Nonlinear-Dynamics-and-Chaos-with-Student-Solutions-Manual-With-Applications/Strogatz/p/book/9780813350844) Additional Resources 3 Geometry of Biological Time - Arthur Winfree 4 Dynamical Systems - Abraham and Shaw 5 Computational Beauty of Nature - Flake	
Assessment	The core assessment is via the group project. Working alone or in pairs students present and analyze a dynamical systems model or technique. Project Presentation (30%): a 10-15' presentation + questions presentation of the project. Project Report (60%): produce a detailed project report as a jupyter notebook. Both the presentation and the report are peer graded according to a rubric. 2 x Brightspace Quiz (10%).	
Location	ErasmusMC	

NB4070	Soft Matter	6
Responsible Instructor	T. Idema	
Contact Hours / Week x/x/x/x	6/6/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Mathematical skills as taught in e.g. the physics or nanobiology BSc programs: linear algebra, calculus, and differential equations. Thermodynamics and statistical physics at the BSc level (including the laws of thermodynamics, concept of free energy, and partition functions).	
Course Contents	Soft matter physics is the study of materials that are 'easily deformable, e.g. by thermal fluctuations or small shear forces. Fluids, gels and biological tissues are important examples. In this course, we study the properties of such materials with a special focus on applications in biophysics and biomaterials. Topics include macroscopic and microscopic fluid dynamics, elasticity, viscoelastic materials, polymers, mixtures, and membranes. For all of these we will build models, using the tools of statistical physics, and apply these models to biological examples, such as DNA (polymers), various tissues (viscoelastic materials), liquid-liquid phase separation, and membrane shapes and interactions. We will introduce the relevant mathematical framework where needed, using ideas from tensor calculus, differential equations and differential geometry. At the end of the course, students read and present a recent research paper on a soft matter topic, connecting what theyve learned to ongoing developments.	
Study Goals	At the completion of this course, the students will be able to: Give an overview of the various topics that constitute soft matter. State, explain and apply the basic laws of fluid dynamics and elasticity theory. Explain what a constitutive equation is and derive such equations for given viscoelastic materials. Mathematically describe polymers and membranes. Mathematically describe multi-component systems. Apply the above to various components of living cells.	
Education Method	Lectures and tutorials (until Christmas), reading, discussing and presenting research papers (in January). Video lectures covering the mathematical derivations, complemented by notes, will be provided each week. On-campus (or, if necessary, live online) sessions with the teachers will complement these lectures, focusing on concepts, with ample room for questions and working out example problems. Students work on problems themselves (with the help of TAs) during tutorials; some of these worked solutions are to be handed in each week and will be graded.	
Literature and Study Materials	Required: Lecture notes & articles provided during the course. Recommended: M. Doi, Soft Matter Physics, Oxford University Press, 2013, ISBN 978-0-19-965295-2. Related literature: D. Boal, Mechanics of the cell (2nd edition), Cambridge University Press, 2012. L. D. Landau and E. M. Lifshitz, Fluid mechanics (2nd edition, 1987) and Theory of elasticity (3rd edition, 1986), Butterworth Heineman. P. M. Chaikin and T. C. Lubensky, Principles of condensed matter physics, Cambridge University Press, 1995.	
Assessment	Homework assignments (problems, 20%), paper presentation (10%), midterm exam (first part of course, 35%), final exam (second part of course, 35%). There will be a single resit which replaces both the midterm and final exam. If there are very few participants for the retake, it may be given as an oral instead of a written exam. There is no retake option for the homework; if students cannot give their presentation for legitimate reasons and inform the teacher in time, they will be given an alternative option to present.	
Remarks	Students interested in learning more about the maths behind polymers and (especially) membranes are encouraged to check out the elective Geometry of physics (NB4110).	
Location	Delft	

NB4510	Proposal Development Part 1	3
Responsible Instructor	Dr. R.J.T.S. Palstra	
Instructor	S. Rao	
Instructor	Dr. W.J. Lans	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	This course is required for NB4520 Project Development part 2. Students must have passed both before beginning their MEP or have special permission of the academic counsellor.	
Course Contents	This course is connected to course NB4520. Together both courses focus on different aspects of developing a research project, such as a literature review, presenting ideas and results and writing a project proposal.	
Study Goals	The study goals of each module of this course are: Research toolbox - skills development module: 1. Reflect on what it means to do scientific research with integrity and recognize signs of scientific misconduct 2. Explain how scientific funding works and learn the basics of what is expected when applying for research funding 3. To create awareness about proper data management 4. To create awareness about valorization and the role of the technology transfer office 5. To perform proper literature searches as well as keeping a bibliographic database Literature review module: Upon completion of this module you will be able to: 1. Write an insightful short scientific review on a specific topic within the field of Nanobiology. 2. Gather literature on a chosen topic, evaluate its usefulness, and extract relevant information. 3. Organize the information into a short scientific review understandable to a general science audience. 4. Write a review in comprehensible and structured written scientific English. 5. Prepare a short review according to a given format, including referencing style and preparation of display items (if used) in the same manner expected for publishing such a review. 6. To constructively incorporate feedback from your (MEP) supervisor to improve the review.	
Education Method	lectures, seminars, group work, self-study, writing assignments	
Assessment	Presentation plan pass/fail Literature plan pass/fail Attendance lectures pass/fail	
Special Information	You are not required to have your supervisor for your MEP identified before the course starts, but you should have a general topic and be in discussions with your potential supervisor.	
Continuing Courses	NB4520 Project Development Part 2, then NB5942 Master End Project	
Location	Erasmus MC, the teacher can be reached at r.palstra@erasmusmc.nl	

NB4520	Proposal Development Part 2	7
Responsible Instructor	Dr. W.J. Lans	
Instructor	Dr. R.J.T.S. Palstra	
Instructor	S. Rao	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Required for	Students must have passed this course before beginning their MEP. Students having completed this will enroll in NB5942 Master End Project for 42 ECs.	
Course Contents	This course is connected to course NB4510. Together both courses focus on different aspects of developing a research project, such as a literature review, presenting ideas and results and writing a project proposal.	
Study Goals	The study goals of each module of this course are: Research presentations module: At the end of this course module, the student will be able to: 1. Design a scientific presentation for a specific audience. 2. Prepare an oral script and visual aids for a presentation for a specific time duration. 3. Organize new ideas or summarize a scientific topic in a concise and informative presentation. 4. Deliver an oral presentation on a chosen topic appropriate to a specified audience. 5. Provide constructive feedback on presentation content and style to peers. 6. Reflect on their own presentation skills and identify strengths and areas of improvement Literature review module: Upon completion of this module you will be able to: 1. Write an insightful essay/short review on a specific topic within the field of Nanobiology. 2. Gather literature on a chosen topic, evaluate its usefulness, and extract relevant information. 3. Organize the information into a short review understandable to a general science audience. 4. Write a review in comprehensible and structured written scientific English. 5. Prepare a short review according to a given format, including referencing style and preparation of display items (if used) in the same manner expected for publishing such a review. 6. To constructively incorporate feedback from your (MEP) supervisor to improve the review. Project proposal writing module: Upon completion of this module you will be able to: 1. Find relevant scientific literature and summarize it in comprehensible and structured written scientific English to provide background information on the topic to be studied in your Nanobiology Master research project. 2. To clearly and convincingly state the main scientific question, and specific objectives, that will be addressed. 3. To identify and advertise the innovative aspects of the research project in written form. 4. Describe a realistic experimental approach and time plan to answer the research question. 5. Identify and convincingly describe the possible knowledge utilisation aspects of the expected results of the research project 6. Combine all the above products into a complete research project proposal such as those prepared in order to obtain funds from granting agencies. 7. To critically evaluate such a proposal from peers and propose improvement during an evaluation meeting. 8. To constructively incorporate feedback to improve the scientific proposal. Research toolbox - skills development module: 1. Reflect on what it means to do scientific research with integrity and recognise signs of scientific misconduct 2. Explain how scientific funding works and learn the basics of what is expected when applying for research funding 3. To create awareness about proper data management 4. To create awareness about valorization and the role of the technology transfer office	
Education Method	lectures, seminars, group work, self-study, writing assignments	
Prerequisites	Completion of NB4510 Project Develpment Part 1.	
Assessment	Presentation counts for 20% endgrade) Literature review counts for 40% endgrade) Project proposal counts for 40% endgrade)	
Special Information	Students must have identified a GreenList supervisor before they begin this course for their potential MEP.	
Location	Erasmus MC, the instructor can be reached at w.lans@erasmusmc.nl	

NB5015	Seminars	2
Responsible Instructor	I. Chaves	
Contact Hours / Week x/x/x/x	1/1/1/1	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	At the beginning of this course in Q1 there will be 3 in person sessions. Attendance is highly encouraged but not mandatory. Session 1: Introductions and expectations, how to find seminars, form groups, what should be included in a report. Session 2: A faculty seminar on a nanobiology topic Session 3: Share and discuss the seminar reports from session 2. Attend seminars and write seminar reports on your own schedule for the rest.	
Study Goals	To develop skills in gathering information and knowledge from the attendance of scientific seminars.	
Education Method	Extracting information from seminars and reporting on them in writing. You will write an one-page report (~500 words) on each of the 12 attended seminars. Illustrations may be included.	
Assessment	A faculty member will read your reports (a grade will be uploaded within 2 weeks of receiving the last report). This is a pass/fail course. The report should summarize the contents of the seminar, and include (at the end) a personal reflection (discuss whether it was interesting, useful, applicable to your current research, whether you could understand what was presented). The reports will be checked for plagiarism. Reports are submitted via Brightspace.	
Special Information	The teacher can be reached at i.chaves@erasmusmc.nl	
Location	Intro sessions are at Erasmus MC.	

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Master Nanobiology**

Electives Nanobiology	
Responsible Program Employee	J.C. Colgrove
Introduction 1	In addition to the obligatory courses in the first and second year, students must earn at least 22 EC of elective courses. Students may request to include electives to their degree audit which are not on the approved list. Students must submit this request to the Board of Examiners before beginning the course(s). Students may take one elective (no more than 6 EC) that is not related to Nanobiology content. It may be taken at any university but must be approved by the Board of Examiners in advance to ensure appropriate level of study.

NB4080	Protein Quality Control Mechanisms	3
Responsible Instructor	Dr.ir. N.J. Galjart	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	While cell identity is encoded in the DNA, it is largely conveyed by proteins, the workhorses of cells. This elective is about the life of proteins, which are dynamic entities produced in a large variety of sizes and shapes so as to fit their different functions. As they are synthesized proteins are folded, often in highly complex manners, so that they can, for example, form multimeric structural components, or single protein enzymatic machines. The proteome of a cell constantly needs to adapt to cues and also risks to be misfolded or damaged, due to ageing or other processes. This can cause protein aggregation, which in turn can result in toxicity and disease. To keep the cell in a good shape, protein synthesis and turnover are therefore highly regulated. Cells contain several protein quality control checkpoints and networks that can resolve damage at different scales, i.e. at the level of individual proteins, at the level of organelles and cells, and even across cells and tissues. These protein quality control mechanisms are also important for biotechnological applications requiring the production of recombinant proteins. The aim of this course is to understand protein synthesis, localization and turnover, and the regulation of proteome health by the different protein quality control networks. The course is given from a cell biological perspective, with a focus on microtubules, an essential cytoskeletal network. The course consists of lectures and class and homework assignments and is concluded with a written exam. During the assignments the students will browse various databases, analyze protein structures using examples from the lectures, perform a number of other practical assignments, and even prepare their own lecture. The course has links with the areas of biochemistry, biophysics, biotechnology, and bio-informatics. Note that presence during the lectures is mandatory and that both the "class and homework assignments" and the "written exam" have to be succesfully completed.	
Study Goals	At the completion of this course, the students will be able to describe: - different types of mammalian RNAs and their functions; - basic steps of mammalian protein synthesis and folding; - protein localization and compartmentalization in mammalian cells; - protein quality control checkpoints and mechanisms. Using this general knowledge the students will be able to assess: - advanced literature covering the subjects listed above; - the function and turnover of the microtubule cytoskeleton.	
Education Method	Lectures and worksessions	
Assessment	- Class and homework assignments: 50%; - Written exam: 50%. Both the "class and homework assignments" and the "written exam" have to be succesfully completed (i.e. 2 x pass). Presence during the lectures is mandatory, both to pass the class and homework assignments and to be able to participate in the written exam. This class is graded on a Pass/Fail basis.	
Special Information	The teacher can be reached at n.galjart@erasmusmc.nl	
Elective	Yes	
Location	Rotterdam EMC	

NB4090	Stem Cells	3
Responsible Instructor	Dr. D. ten Berge	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Biology of Cancer NB4040.	
Course Contents	This course is about stem cells, their biology, their roles in our bodies, and their uses. Stem cells are the building blocks of our bodies. Every cell in our body is either derived from a stem cell or is a stem cell itself, hidden in dedicated niches and ready to generate new cells to replace those that are lost or damaged. This course will present the most important milestones and current developments in stem cell research. Among the topics will be pluripotent stem cells, cloning and reprogramming, chimeras, embryology, differentiation and self-organization of stem cells into tissues and organs, adult and organ stem cells, stem cell niches and signaling pathways that regulate stem cells, disease modelling using stem cells, organoids and organ-on-chip models, and clinical and diagnostic applications of stem cells. The course has a strong focus on embryology and biology of stem cells. Note: this course contains some overlap with the Bachelor Medicine minor Biomedical Research in Practice.	
Study Goals	At the completion of this course, the students will be able to: Demonstrate an understanding of the characteristics of stem cells, their microenvironment, the major signaling pathways that regulate stem cells, and their therapeutic potential. Demonstrate understanding of concepts of developmental potency, cloning, reprogramming, chimeras, lineage tracing, differentiation, self-renewal and self-organization. Describe experimental approaches to demonstrate the presence of stem cells. Demonstrate awareness of ethical issues and policy questions with human embryos and stem cells. Describe the rationale behind disease modelling and regenerative medicine approaches to disease treatment. Describe clinical problems for which stem cells can provide novel regenerative therapies. Design stem cell-based approaches to disease treatment. Demonstrate ability to critically read, interpret, summarize and present, together with fellow students, recent scientific journal articles about stem cells.	
Education Method	The course includes the following eduction methods: -interactive lectures -workgroups -journal article reading assignments -group assignments including a presentation of a journal article -instruction in small groups about reading and interpreting scientific papers The exact format may change and lectures and assignments may move online due to possible coronavirus regulations. Students will gain experience in the following professional skills: -reading scientific literature -preparing and holding a group presentation -team work	
Literature and Study Materials	Molecular Biology of the Cell, 6th edition, Alberts et al, Garland Science. Chapter 22.	
Prerequisites	Student must have passed Biology of Cancer NB4040, or have a biology or medical background.	
Assessment	Please note: attendance of all sessions is required. The assessment consists of two parts: a written exam containing open and multiple choice questions, and a journal club presentation. Re-exam is only possible for the written exam. The re-exam can also be an oral exam, to the discretion of the teacher. The final grade is composed of ¾ of the exam and ¼ of the journal club presentation. Exam and journal club presentation can compensate for each other. A final grade will not be awarded if a student has missed any session, unless this was due to circumstances outside the students control. In that case the student needs to contact the teacher for an alternative assignment. Registration for exams is done via Osiris.	
Special Information	Teacher can be reached at d.tenberge@erasmusmc.nl	
Elective	Yes	
Studyload/Week	3 ECTS = 84h 4h/week * 8 weeks = 32h lectures and class leaves 5h/week study time at home and 12h exam preparation Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Rotterdam	

NB4100	Nuclear Architecture	3
Responsible Instructor	Dr. K.S. Wendt	
Co-responsible for assignments	Dr. M.W.J. Fornerod	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The cell nucleus is a storage and reading device for the genetic information where also information on cell fate decisions is stored in the form of switches that turn genes on or off. For example, cellular decisions such as differentiation from a stem cell into a different cell type, but also pathologies like the uncontrolled cell division of cancer cells leave their mark on the genome in the form of characteristic binding of transcriptional regulators and histone modifications. In this course we will discuss: Functions of the main nuclear components (DNA, RNA, nucleoli, bodies and speckles, nuclear membrane) How is the genetic information organized in the cell nucleus (from histones to higher order chromatin organisation, special DNA regions like centromeres and telomeres)? How is chromatin manipulated by regulators, like remodelers and transcription factors, to activate and silence genes? How does the nuclear interior communicate with the cytoplasm (nuclear pores and nuclear transport)? Key methods to study nuclear processes For a preview of the course content you can find links to Youtube animations in the course content section on Brightspace.	
Study Goals	Learning goals of the course: - You will be able to describe the basic principles of gene regulation, chromatin organization and chromatin re-organizing processes. - You will be able to relate the functions of different nuclear components (for example the nuclear lamina) with gene regulation. - You will be able to describe the basic principles of nuclear transport. - You will be able to explain which role phase separation might play for nuclear processes. - You will be able to describe at least one human pathogenic condition that is related to structural malfunctions in the cell nucleus. - You will be able to interpret experimental results obtained with methods and technologies to study nuclear processes and be able to select appropriate techniques to address questions around the major principles discussed. - You will improve your skills to critical evaluate scientific literature and discuss your evaluation in the class.	
Education Method	Course structure: Each course day consists of a lecture and a work group assignments. These are literature discussions, here we will discuss scientific publications that, for example are supporting or contradicting a certain finding (example 30 nm fiber) or discuss novel emerging concepts (example phase separation). Depending on the number of attendants, groups of 2-4 students will be formed that present a study pro and contra while the remaining students critically discuss the presented arguments. Further groups will work on assignments around the interpretation of genomics data related to course content using the UCSC genome browser. Throughout the course we will use Mentimeter for small quizzes. Note that this is the course structure for teaching in a classroom. There might be some changes to the structure in case the course needs to be an online course.	
Literature and Study Materials	Literature: The main reading textbook is the latest edition of the Molecular Biology of the Cell. Complementary articles will be provided in the course content pages on Brightspace. For the literature discussion, all necessary publications together with a small description of the discussion topic will be available on Brightspace.	
Assessment	The course will be assessed by an exam with open and multiple-choice questions. The literature discussion as well as the UCSC browser assignment will be graded as pass/fail and will be together honored in the exam with 30% of the exam points. This applies also to an eventual resit.	
Enrolment / Application	Enrol in the course in Brightspace at least 1 week before the first day of the course so that literature discussion assignments can be prepared matching to the number of students.	
Special Information	The teacher can be reached at k.wendt@erasmusmc.nl	
Location	Erasmus MC	

NB4110	Geometry of Physics	6
Responsible Instructor	T. Idema	
Contact Hours / Week x/x/x/x	0/1/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	BSc-level mathematics and physics, particularly calculus, linear algebra, differential equations, and mechanics. Prior knowledge of special relativity will be helpful (will be repeated / covered at the start of the course), as well as some familiarity with continuum dynamics (e.g. hydrodynamics) and tensors. Most participants also take (or have taken) NB4070 Soft matter / AP3511 Biophysics, which is helpful but not required.	
Course Contents	In this advanced elective the aim is to provide students with the necessary mathematical tools from tensor calculus and differential geometry to be able to describe physical systems where geometry is important. Two important applications are (bio)membranes as two-dimensional surfaces living in three-dimensional space, and the four-dimensional spacetime of general relativity. This is a 'reading course', meaning that the students study the relevant material by themselves, work on the associated problems together, and meet once a week with the instructor to discuss their progress and ask any questions they have.	
Study Goals	At the completion of this course, the students will be able to: - Explain what tensors are, and use them in calculations, both in index notation and index-free form. Describe the geometry of objects in any dimension with the curvature tensor. Explain where the various terms in the Canham-Helfrich energy of the biological membrane come from, and use this energy to calculate membrane shapes. Explain where the various terms in the Einstein equation of general relativity come from, and use this equation to calculate the geodesics of stars and black holes. Use the large-scale description of the universe to discuss its possible evolution scenarios.	
Education Method	This is a 'reading course', meaning that the students study the relevant material by themselves, work on the associated problems together, and meet once a week with the instructor to discuss their progress and ask any questions they have. At the end of the course, all students will independently study an additional topic which they will present to their peers and the instructor.	
Literature and Study Materials	B. Schutz, A first course in general relativity, second edition (2009), Cambridge University Press, ISBN 978-0-521-88705-2. Lecture notes provided during the course.	
Assessment	Problems (discussed in meetings with the other students and the instructor), final presentation. Grades will be based on active participation and on the final presentation (content and delivery both), and determined as a combination of peer grading and the assessment of the instructor. A rubric for peer grading will be provided to the students. Due to the nature of the assessment, there is no resit for this course.	
Special Information	This is typically a very small course, with 5-6 students participating. The minimum number of participants is four; we could accommodate up to ten students in two study groups. The course is open to all interested MSc students from any programme (at TU Delft or elsewhere in the Netherlands), given that they have the necessary background. PhD students are welcome to participate; Delft/Leiden PhD students in the Casimir research school can earn graduate school credits with this course.	
Elective	Yes	
Tags	Challenging Complex Mathematics Intensive Involved Mechanics Small groups	
Location	TU Delft	

NB4120	Biological Networks; a data driven approach to discovery and understanding	3
Responsible Instructor	Dr. M.W.J. Fornerod	
Instructor	Dr. I.V. Smal	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The concept of networks is broad and general, and it describes how things are connected to each other. Networks are present in every aspect of life. In this course the students will learn the essential aspects of networking, and how to apply them to biological data. This includes visualization and discovery in protein, signal transduction and genetic networks, and use these networks to compare them to biological or clinical behavior, such as drug response or patient's survival. Topics covered are network theory, network metrics, use of deep learning networks and chemical reaction circuit design and simulation. Computational tools include Python, Cytoscape with various plugins to analyze large data sets and R packages. Course will run for two months with one session of 3 hours per week. In each session, students will present the result of their previous week's network analysis. After that, an introductory lecture is given, followed by a computer practical, which is continued as assignment.	
Study Goals	In this course the students will learn the essential aspects of networking, and how to apply them to biological data. This includes visualization and discovery in protein, signal transduction and genetic networks, deep learning, network design and use these networks to compare them to biological or clinical behavior, such as drug response or patient's survival.	
Education Method	Course will run for two months with a weekly session of 3 hours. In each session, students will present the result of their previous week's network analysis. After that, an introductory lecture is given, followed by a computer practical, which is continued as assignment. The techniques used in the computer assignments will be examined in the assignment part of the exam.	
Assessment	The course will be assessed by a written exam (closed book) and computer exercises using techniques from the practicals using the participants' own laptop computer.	
Location	Erasmus MC	

NB4150	The Origin and Synthesis of Life	6

Responsible Instructor	Dr. C.J.A. Danelon
Contact Hours / Week x/x/x/x	Not offered in 2022-2023
Education Period	Different, to be announced
Exam Period	none
Course Language	English
Expected prior knowledge	Students should have followed a course in chemistry, biochemistry or biophysics.
Course Contents	Course not taught in 2022-2023, may be permanently cancelled. Part 1: Introduction - Historical survey and the many definitions of life - Conceptual frameworks of research on the origin and synthesis of life - Main scenarios for the emergence of cellular life - Basics on the structure and organization of cells Part2: From prebiotic chemistry to self-replication - Early Earth geochemistry - Prebiotic synthesis of ribonucleotides, amino acids and lipids - Self-organization, self-assembly - Autocatalysis and self-replication - Assembly of (bio)chemical reaction networks Part3: Protocells - Chemistry and physics of primitive membranes - Models of protocells - The RNA World and the RNA protocell - Emergence and biological complexity Part4: Minimal cell and synthetic cell - Engineering minimal cells, bottom-up vs top-down - Scaffolds for a synthetic cell - Cell-free gene expression - In vitro evolution - Social and ethical issues of proto- and artificial cells
Study Goals	To have knowledge about prebiotic processes, self-organization, autocatalysis, compartmentalization, as well as their mutual interactions and possible implications in the transition to life. To be able to formulate the pros and cons of the different existing scenarios for the emergence of complex biomolecules and protocells. To be able to formulate the main challenges towards the realization of a synthetic cell and to propose possible solutions. To understand international research literature related to the theory covered. To be able to write a summary of the studied article(s) as well as a critical discussion and possible follow-up research. To present a research paper from the recent literature in a clear and interesting manner, meeting the requirements of scientific communication. To answer questions and provide detailed argumentation about the article(s) presented as indicated above.
Education Method	Lectures Article discussion/presentation Small report writing Interactive sessions Homework
Assessment	Final written exam (50%) Oral presentation & written report (30%) Homework & short in-class assignments (20%) Important: students can resit the final written exam only, not the presentation/report and homework.
Permitted Materials during Tests	None, unless communicated otherwise during the course.
Remarks	Course not taught in 2022-2023, maybe permanently cancelled.
Elective	Yes

NB4160	Engineering of living systems	3
Responsible Instructor	Dr. G.E. Bokinsky	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	This course is intended for students from Nanobiology; however students from other programs are certainly welcome. Prior knowledge of molecular biology at a basic level is assumed. Material for students with minimal biology background can be provided.	
Course Contents	Even the simplest organism is more complex than the most sophisticated man-made device. How did biological evolution build such intricate machines, and could we build such organisms ourselves? In this course, we will examine how to engineer the properties of existing organisms. In doing so, we will consider the both the successes and failures of modern synthetic biology in its efforts to apply general engineering principles to biology. Through lectures and interactive discussion of recent research articles, we will develop an integrated view of biological behavior, evolution, and the state-of-the-art of bio-engineering. We will explore how we can use modern genetic engineering tools to engineer the properties of organisms. We will study basic tools and advanced synthetic-biology approaches that can be used to engineer biological solutions for real-world problems. A background in biology will be appropriate preparation for participating students. Throughout the course you will get the opportunity to study, present, and discuss recent articles from the scientific literature on cutting-edge discoveries and techniques. Specific topics include: real-time experimental evolution, the modern genetic toolbox, feedback control and logic gates, design properties and construction principles of synthetic biology circuits and networks, and the debugging and optimization of engineered networks.	
Study Goals	By the end of the course, students will be able to: -Summarize how living systems work in terms of structural and functional concepts at multiple levels of biological organization. -Identify why living systems are engineerable. -Describe the limitations of the use of engineering approaches learned from other fields when applied to biology. -Apply design principles to create new biological functions. -Improve and debug the results of genetic engineering by utilizing specific genetic and evolutionary principles. -Critically evaluate literature papers in the fields of synthetic biology. -Assess the social and safety implications of genetic engineering and directed evolution projects. -Engage in constructive scientific discussions	
Education Method	The course will be conducted using a combination of styles to promote student involvement and interaction with the material. Lectures describing and explaining the topics of study will be supplemented with readings from ground-breaking journal articles. Students will present and lead discussions about research articles.	
Literature and Study Materials		
Assessment	Students will be assessed based upon the following: * Literature case study: oral presentation of a journal article and leading a classroom discussion that follows (40%) * Written reflection on the impact of a journal article assigned at the end of the course (30%) * Written responses to journal articles and participation in classroom discussion (30%) Retake possibility: Students who do not receive a passing grade on the Written Reflection assigned at the end of the course are given one opportunity to submit a reflection on a second article.	
Permitted Materials during Tests		
Elective	Yes	
Location	Delft	

NB4165	Molecular Virology & Immunology	3
Responsible Instructor	Dr.ir. S.J.J. Brouns	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Molecular biology at the BSc level Cell biology at the BSc level Microbiology at the BSc level	
Course Contents	The importance of viruses for the evolution of life is paramount. The recent pandemic caused my SARS-Cov-2 has underlined the impact viruses can have on all areas of human society. In this course we will discuss the principles that viruses employ to infect bacteria, archaea and eukaryotes. We will discuss how viruses evolve and look at the main mechanisms of immune defense of hosts, including CRISPR-Cas, adaptive immunity and RNAi. We will discuss bacterial, archaeal and eukaryotic viruses through a series of lectures, worksessions and group presentations. We will invite leading virologists to bring in cutting edge insights into virus biology.	
Study Goals	At the completion of this course, the students will be able to: Give an overview of how various viruses infect their hosts Explain molecular mechanisms involved in distinct steps in virus reproduction/life cycles, such as virus attachment, entry, replication Explain molecular mechanisms of host immunity and immune evasion by viruses Critically select, review, discuss and present research papers in virology Formulate relevant research questions in virology	
Education Method	Lectures, discussion groups, student presentations	
Literature and Study Materials	Required: Lecture notes & articles provided during the course.	
Books	Recommended: Flint, Racaniello, Rall, Hatziioannou, Skalka. Principles of Virology 5th edition, 2022. EISBN13: 9781683673583 Provided digitally Flint, Racaniello, Rall, Skalka. Principles of Virology 4th edition. 2015.	
Assessment	final written exam, student presentations	
Location	TU Delft	

NB5910	Independent Research	0
Responsible Instructor	T. Idema	
Course Coordinator	J.C. Colgrove	
Contact Hours / Week x/x/x/x	variable	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Interdisciplinary independent research on current topics with scientific or societal relevance.	
Study Goals	At the end of the project, students will have learned to: - develop feasible research questions - apply innovative methods to the address these questions - communicate about scientific work in an effective way - write reports about their findings	
Education Method	In this course you carry out an independent research project with the help of a scientific advisor from the program. If you would like to do such a project, you may join an existing project (an example would be the Corona Research super project spring of 2020) or propose one to the Programme director. Activities could include literature review, modeling, and lab work, and have to include scientific discussion and report writing.	
Assessment	Grading is Pass/Fail. You will be assessed on your research log of work done, communication with your supervisor and other members of your research group, and your final written report.	
Special Information	Students wishing to initiate an independent research project must send a proposal to the programme director and have it approved before beginning the project. The proposal should include a description of the research project, a time planning, a supervision plan (e.g. frequency of meetings with supervisor), deliverables (which will be assessed), and the number of EC the student will receive upon successful completion. The proposal should be coordinated with and approved by the intended supervisor before submission to the program director.	
Elective	Yes	
Studyload/Week	Credits are variable, students receive up to 1 EC per week based on at least 28 hours per week, with fractions thereof for less weekly effort. The total amount of credits given cannot exceed the amounts of credits indicated in the project proposal. Credit requests with justification (log book and final report) must be sent to the programme director who will assess the effort and register credits in Osiris.	

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Master Nanobiology**

Other Electives Courses 2022	
Responsible Program Employee	J.C. Colgrove
Introduction 1	In addition to the obligatory courses in the first and second year, students must earn at least 22 EC of elective courses. There are several courses offered by the Nanobiology programme. In addition, the board of examiners has approved a substantial list of courses from other programmes which students may include on their elective list. Students may request to include electives to their degree audit which are not on the approved list. Students must submit this request to the Board of Examiners before beginning the courses. Students may take one elective (no more than 6 EC) that is not related to Nanobiology content. It may be taken at any university but must be approved by the Board of Examiners to ensure appropriate level of study. A limited number of courses from University of Leiden have been approved by the Board of Examiners. They are: 4373MUBI6 Multiscale Mathematical Biology (Leiden University) 4423CHEIM Chemical Immunology (Leiden University)

AP3021	Advanced Statistical Mechanics	6
Responsible Instructor	Prof.dr. J.M. Thijssen	
Instructor	T. van der Sar	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Bachelor (applied) physics or equivalent.	
Course Contents	This course covers the theory of statistical mechanics at the intermediate level, in particular: 1. The formalism of statistical mechanics and its relation to thermodynamics, Ensemble theory 2. Systems of non-interacting particles - classical, quantum 3. Classical systems of interacting particles - perturbation theory: Cluster expansion - Phase transitions: van der Waals, mean field theory, exact solutions - Phase transitions: scaling and renormalization 4. Fluctuations	
Study Goals	Course goals are: 1. Students are able to predict the behaviour of macroscopic, in-equilibrium systems composed of non-interacting particles using the formalism of statistical mechanics. 2. Students are able to analyze the behaviour of macroscopic, in-equilibrium systems composed of interacting particles using various approximation methods of statistical mechanics. 3. Students are able to analyze fluctuations of in-equilibrium systems, as well as the processes by which out-of-equilibrium systems evolve towards equilibrium.	
Education Method	Students prepare the material for each week before the first lecture. For this, they use lecture notes and books, and youtube videos developed by the lecturer. Some control questions requiring simple calculations or the interpretation of particular results are part of the preparation. During the first lecture of the week, the material will sometimes be summarized and students are invited to ask questions on points they did not yet fully grasp. The remaining time of the four hours per week, the students work on problems in groups, while the lecturer and the TA's provide help. Sometimes, some difficult points will be discussed by the whole group. The aim is that every student knows how to finish the problem at the end of these exercise classes for the week. The student can then work out a detailed solution at home and hand this in. The homework which is handed in counts to the final grade.	
Literature and Study Materials	The course follows Lecture notes on Statistical Mechanics by Jos Thijssen From the lecture notes, youtube videos with explanations on many parts of the lecture notes are accessible by mouse click or reading a QR code They can be found on brightspace. Additional textbooks: Statistical Mechanics in a Nutshell, by L. Peliti; Princeton University Press (2011). Other useful textbooks: M. Kardar: Statistical Mechanics of Particles || Cambridge University Press, 2007 Statistical Mechanics of Fields || Cambridge University Press, 2007	
Assessment	The course puts strong emphasis on problem solving skills. Homework is an important element of the course. The grade for the homework can contribute a maximum of 1.5 to the final grade according to the formula: R = (1-0.015*H)*E + 0.15*H where R is the final grade, E is the result for the exam (which may include the midterm, see below) and H is the result for the homework. There is a midterm exam. Students having taken the midterm exam can take a final on the second part of the course material only. There is also an option to take a final exam on all the course material (for students missing the midterm or scoring a bad mark there). If a student chooses to do the final exam on the second part of the material, the exam grade E is calculated according to E = 0.35 M + 0.65 F, where M is the grade for the midterm and F is the result of the final exam. The exam grade E for a student who has decided to the final exam on all the material is just the result of that final exam F: E = F The bonus based on the problems keeps its validity for the retake following the semester in which the course was taught. The score for the problems is reset to 0 for the next time the course is taught. There is a retake in the spring. This retake covers all material, i.e. it is a 'full' exam.	
Permitted Materials during Tests	closed book exam. (Non-graphical/non programmable) calculator allowed.	

AP3032	Continuum Physics	6
Responsible Instructor	Prof.dr.ir. C.R. Kleijn	
Instructor	Dr.ing. S. Kenjeres	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	undergraduate calculus, differential equsations and transport phenomena/fluid dynamics.	
Course Contents	The continuum description of the mechanics of solids and liquids is discussed in a unified approach using tensor analysis. Deformations of a body are studied and the strain tensor and the rate-of-deformation tensor are introduced. This is followed by a discussion of the stress tensor, and the equations of motion of a body are derived based on the balances of linear and angular momentum. For linear elastic solids the following topics are discussed: constitutive equation, Navier equations, energy principles. For Newtonian fluids we discuss the stress tensor, Navier-Stokes equations, and the first and second law of thermodynamics. The lecture series is then continued with a number of special topics a.o. multicomponent reacting flow (combustion), thermal convection and magneto-hydrodynamics (MHD).	
Study Goals	Being able to formulate and apply conservation laws (mass, momentum, energy) and constitutive equations (stress-strain relations, equations of state,) for linear elastic solids and Newtonian fluids. Being able to apply tensor algebra and tensor calculus in the formulation and analysis of laws describing continuum mechanical systems. Having knowledge of the relevant phenomena in multiphysics continuum mechanical systems (reacting flows, thermal convection, magnetohydrodynamics). Being able to solve simple problems in continuum mechanics (kinematics, kinetics, linear elasticity, Newtonian fluid mechanics, special topics covered in the course). Being able to identify some relevant aspects of physics of continuous media in practical situations (relevant phenomena, dimensionless numbers, regimes, simple estimates).	
Education Method	Theory presented in lectures. Practice sessions to make exercises. Assignments to be made at home.	
Literature and Study Materials	The general part of the course is based on the following book: J.N. Reddy, Principles of Continuum Mechanics, 2nd edition, Cambridge University Press, 2018, ISBN: 978-1-107-19920-0 (hardback) For the lectures on special topics (reactive flows, heat transfer and thermal convection, magnetohydrodynamics), slides and/or notes are made available. Additional textbooks of interest: - For the lectures on continuum mechanics: J. N. Reddy, An Introduction to Continuum Mechanics Cambridge University Press; October 2007 ISBN 9780511473883 - For the lectures on reactive flows: K. K. Kuo, Principles of Combustion, 2nd Edition, John Wiley and sons, ISBN 10: 0471046892 and ISBN 13: 9780471046899 - For the lectures on Newtonian fluid mechanics: P.K. Kundu, I.A. Cohen and D.R. Dowling, Fluid Mechanics, 5th Edition, Academic press, 2012,	
Assessment	On-campus or Online exam at end of the second period. Final mark partially (6x2.5%) based on the performance in 6 optional homework assignments, if the assignment grade is higher than the exam grade. Resit exam at the end of the third period. Homework assignments are *not* valid for the resit exam, and also not for next year exams.	
Permitted Materials during Tests	Closed book on-campus or Open book online exam (depending on Covid-19 situation)	
Studyload/Week	12 hours per week (4 scheduled, 8 homework) during 14 weeks.	

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morpological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 6 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: ((0.85*EX + 0.15)+ AS)/ 2 or without the bonus point from the practicals: (EX+AS)/2 With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

AP3162	Physics of Biological Systems: Mathematical Modelling in Systems Biology	6
Responsible Instructor	Dr.ir. L. Laan	
Instructor	Dr.ir. S.J. Tans	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basic differential equations. Not necessary to have any prior knowledge of biology.	
Course Contents	This is a course intended for physicists, biologists, engineers, and quantitative-minded students who are fascinated by the questions: "What is life?" and "What are the physical and mathematical rules that govern living cells and tissues?" This course takes a quantitative approach to understand genetic circuits, pattern formation, and molecular recogniton systems. If you are a physicist and have wondered why so many physicists are entering biology, this course will hopefully answer your question in depth. If you are a biologist and have wanted to study biology with mathematical models and quantitative data, then this course will hopefully teach you how to do so. In recent years there has been a tremendous progress in revealing the fascinating internal dynamics of cells. The driving promise is to understand living systems bottom-up, by combining existing knowledge of individual proteins and reactions, novel techniques to follow single cells and molecules in time, and simple mathematical models to explain the organisation and dynamics. In this course we will focus on the key innovative concepts that have been discovered using model systems. The course aims to cover the following topics: basic biology, pattern formation inside cells, gene regulation, temporal and spatial dynamics, stochasticity and noise, evolution, genotype to function relations and cytoskeletal dynamics Students will also explore results of cutting edge, modern research, where novel experimental or theoretical approaches are used to investigate central questions in biology. By reading and discussing research papers that are just months or 1-2 years old, the students will get to directly apply their knowledge from this course to research papers in this fast moving field driven by engineers, physicists, and quantitative biologists.	
Study Goals	- To be knowledgeable about how math and physics can be applied to study cellular processes. - To be knowledgeable about modelling the following key concepts:, pattern formation, stochasticity and noise in cells, evolutionary dynamics, temporal and spatial organization of cellular processes, cytoskeletal dynamics. - To be able to design and interpret a quantitative experiment to address concepts listed above. - To be knowledgeable about current research literature in quantitative biology and physics of life.	
Education Method	Lectures and one-on-one discussions with the instructors and TAs Students presentations	
Literature and Study Materials	We will provide the study material in class.	
Assessment	The final grade for the course will be determined as follows: - report about a paper (30%) - presentation about a paper (30%) - Final exam (40%) In order to pass this course, students must obtain at least a passing grade (5,8) on the final exam and an overall grade of at least 5,8.	
Permitted Materials during Tests	all material handed out to you during the course	
Elective	Yes	

AP3232	Medical Imaging Signals and Systems	6
Responsible Instructor	Dr.ir. M.C. Goorden	
Instructor	Dr. D. Maresca	
Instructor	Dr. S.D. Weingärtner	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Elementary physics, particularly related to signal analysis.	
Course Contents	The course Medical Imaging Signals and Systems covers the principles of advanced medical imaging modalities. Particularly, students will study the physics, acquisition techniques and signal processing underlying conventional X-ray, Computerized Tomography (CT), Single Photon Emission Computed Tomography (SPECT), Positron Emission Tomography (PET), Ultrasound (US) and Magnetic Resonance Imaging (MRI). Course Content: Introductory lecture with the history of medical imaging and its importance for modern medicine and biomedical research. Conventional X-ray, CT, SPECT and PET imaging (33% of content): Basic principles of planar X-ray imaging and scintigraphy (short); Advanced methods for image acquisition using CT, SPECT and PET; Mathematical techniques for image reconstruction, including filtered backprojection and iterative approaches; Issues of image quality, especially related to resolution, signal to noise ratio, quantitative and dynamic imaging. MR imaging (33% of content): Fundamentals of MRI: spin, Larmor frequency, T1, T2, T2* contrast (short); Spatial encoding: K-space; Image Quality: PSF, signal strength and noise level; Advanced MRI techniques such as diffusion, perfusion and flow imaging. Ultrasound imaging (33% of content): Physical and mathematical principles of ultrasound imaging, including probe design and advanced US imaging techniques such as harmonic imaging and non-linear reconstruction techniques for imaging tissue properties. The course will end with a mini-symposium on specialized imaging techniques. Therefore, recently developed medical imaging methods will be studied, going beyond the theory of the lectures. Students working in pairs will write a short essay on a selected technique and present it to the group.	
Study Goals	1. Acquire in-depth knowledge about the physics and image reconstruction underlying X-ray, CT, acoustical and magnetic resonance imaging; 2. Being able to solve elementary problems related to the theory mentioned in 1; 3. Being able to solve more advanced problems addressing the theory mentioned in 1 by combining mathematical skills and physical insight; 4. Able to acquire new knowledge about clinical a medical imaging technique; 5. Being able to present newly acquired knowledge about a medical imaging technique.	
Education Method	Lectures and an assignment.	
Literature and Study Materials	Book: Medical Imaging Signals & Systems, Jerry L. Prince, Jonathan Links. Upper Saddle River NJ: Prentice Hall, second edition, 2014, 544 pp, ISBN-13: 978-0132145183. Additional handouts wherever necessary.	
Assessment	The course on Medical Imaging comes with an assignment for which you will need to hand in a paper and make a presentation. This is an obligatory part, for which you will get a grade from 1-10 (A). Additionally, there will be an exam also graded from 1-10 (B). The grades for both parts needs to be at least 6.0 in order to pass. The final grade is calculated by: (A + B)/2. It is not possible to take part in the exam if you have not made the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Pocket calculator.	
Elective	Yes	

AP3371	Radiological Health Physics	6
Responsible Instructor	D.S. Dihalu	
Instructor	M. Schouwenburg	
Instructor	Dr. K.R. Huitema	
Contact Hours / Week x/x/x/x	0/0/8/8 (friday) = Note that the course starts earlier and ends later than Q3.	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge		
Summary	The course provides a broad overview of modern health physics. All subjects of health physics will be discussed in depth during a series of lectures. The course also consists of a practical part which is mandatory. The final exam is a theoretic exam in which both the student's ready knowledge and problem solving skills in the field of radiation protection are tested. This course brings participants up to the level of allround radiation protection experts, who may function as a coordination expert in a facility. With sufficient results on the final exam you receive an additial nationally recognized diploma "Coordinating Expert in Radiation Protection". The diploma is required for coordinating radiation experts, but also experts in radionuclide laboratories, medical clinical physicists and doctors of nuclear medicine need the diploma for their work.	
Course Contents	Lectures and tutorials on: - Radiation, radioactivity, decay. - Radiation sources. - Interaction of radiation with matter. - Methods of radiation detection. - Radiation dosimetry. - Radiation shielding. - Biological effects of ionizing radiation. - Internal dosimetry. - Natural and man-made sources/background radiation. - Radiation protection philosophy (protection principles). - Rules and regulations; organizational, procedural and administrative tasks. - Safety measures; operational radiation protection. - Radiation protection when handling open sources. Mandatory parts: - Practical experiments. These include prior preparation at home, active participation during the experiments, documenting results, reporting including discussion and conclusion/giving advice. All these aspect of the practical work will be monitored and scored during the course. - Several written assignments during the course. - Special modules on: Laser Safety, Communication, Organization & Task and Responsibilities, Risk Analysis and Ethics.	
Study Goals	This course will teach you how to work safely with sources of ionizing radiation. It is a broad introduction into the field of radiation protection and the application of ionizing radiation in science, medicine and industry. Because of this broad coverage of the subject it consists of theoretical parts, a mandatory practical part and special modules on topics like laser safety, communication, organisational aspects and risk analyses.	
Education Method	Oral lectures (classes), tutorials/instructions, assignments, lab experiments	
Literature and Study Materials	Massiot P, Jimonet C, European Radiation Protection Course: Basics. EDP Science (2014), ISBN: 978-2-7598-0703-1. Can be purchased on day 1 or 2 of the course.	
Practical Guide	The practical workbook will be made available on Brightspace on the second day of the course.	
Assessment	National exam in May. Resit in December. Two parts: - part 1: Multiple Choice; written closed book exam (max 33 points) - part 2: 4 problems; essay type questions, written open book exam (max 67 points)	
Permitted Materials during Tests	Part 1 (Multiple Choice part): non-programmable calculator and 'clean' dictionary. Part 2 (Essay type questions): All course materials like the textbook, notes, hand outs, etc.	
Special Information	For students from other universities: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student/	
Elective	Yes	
Studyload/Week	1 day/week classes + 0.5 day/week labs + 0.5 day/week preparation	
Schedule	Note that the course starts earlier and ends later than Q3. The detailed course program will be published on Brightspace in the second week of January.	
Location	Reactor Institute Delft (building 50) Mekelweg 15 NL - 2629 JB DELFT	

AP3582	Medical Physics of Photon and Proton Therapy	6
Responsible Instructor	Z. Perko	
Instructor	Dr.ir. D. Lathouwers	
Instructor	Prof.dr. M.S. Hoogeman	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Some knowledge in basic programming with Python is required. The assignments can be made in any program that you are familiar with, but the big assignment notebooks are provided in only Python.	
Course Contents	The course deals with various topics concerning radiotherapy (mainly photon and proton): Introduction to radiotherapy Clinical aspects Radiobiology Biophysical models Proton therapy physics Dose shaping and plan optimization Dose calculation using clinical and Monte Carlo schemes Theory of dosimetry and in-vivo dosimetry Brachytherapy	
Study Goals	Main learning objectives 1. The student can define and explain the main clinical aspects of radiotherapy for different treatment modalities, list and describe the steps of a typical radiotherapy workflow. 2. The student can list and describe the basic principles of radiobiology and discuss their practical consequences for radiotherapy. 3. The student can recall and explain the function of biophysical models in radiotherapy, is able to apply these models and evaluate different treatment approaches by interpreting results. 4. The student can describe the main parts of the machinery used in various modes of radiotherapy and explain their role in administering dose. 5. The student can explain the clinical treatment planning process, can apply dose optimization methods and solve simplified planning problems, and can create and evaluate different treatment plans for idealistic model geometries. 6. The student can identify possible sources of errors arising in radiotherapy, and explain and describe correction methods used in the planning process and in practice for high precision radiotherapy. 7. The student can describe and explain the basic physics related to particle acceleration and beam transport, and can perform dose distribution calculations.	
Education Method	Lectures	
Literature and Study Materials	Handouts of powerpoint presentations and questions	
Assessment	Written examination (oral when below certain threshold) and assignments during course. The assignments are mandatory. Resit exam can be oral (by appointment), depending on the number of students.	
Permitted Materials during Tests	Non-graphical/non-programmable pocket calculator.	
Elective	Yes	
Tags	Physics	

AP3831	Systems Engineering for Physicists	3
Responsible Instructor	Dr. M. Sovago	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This course is an obligatory preparatory course for the Systems Engineering (Physics Design) Company Project (AP3841). TU Delft Master's students at Applied Sciences (and EEMCS) Faculty are welcome to join the course. It is meant for the first year Master's students.	
Course Contents	During Systems Engineering (SE) course for applied scientists, you will learn and discover the art behind designing, building and managing complex systems. You will get to master SE tools and methodologies as well as management capabilities. Every week you will learn the theoretical aspects of SE during an interactive lecture. Step-by-step you have the possibilities to discover the process and the tools that we can use to design a complex system. You can apply the knowledge from the lecture to design your own system. A few systems will be a-priory defined by the lecturer. You can the possibility to indicate your top 2 preferences. Together with a team of ~ 5 peer-students, you will work together for 10 group project ses-sions on one of the systems: 1. Define the problem you will solve: Problem Statement 2. Define the requirements of the system: Requirements specification 3. Interview possible stakeholders 4. Define the functions of the system: Functional Decomposition 5. Discover possibilities to fulfil the requirement: Means and Morphological Analysis 6. Model and Design your System 7. Create a test planning: fulfil the system functions, plan a test on paper and possibly try it in the lab We do not aim to build a prototype of the designed system, but possible if system complexity allows it. This SE process will be implemented to design and build a complex system during the AP3841 Systems Engineering company project (2nd year Master).	
Study Goals	By the end of this course, you will be able to: 1. Recognize the need and relevance of Systems Engineering (SE) for every applied scientist 2. Structure the key steps in Systems/Product/Process Development 3. Systematically breakdown tasks and work in interdisciplinary/multicultural teams to apply and design a complex system 4. Apply SE methodologies and SE tools in projects 5. Learn and apply transferable skills along the course: chair a team for ~2 weeks, team management and team roles, presentation and writing skills (based on weekly activities), self- and team-reflection.	
Education Method	The course is centred around 10 group project sessions. You will work in a group of about 5 students with different backgrounds and study tracks. Each session focuses on one aspect of Systems Engineering. Weekly group project cycle: 1. A plenary and interactive lecture: learn the SE tools and methodologies. You will be able to ask questions and have discussions with your lecturer 2. Team activity: you will implement SE tools to design and engineer a pre-defined system. You will be working with your peer-students and be guided by your TA. Each week you will have a chair (a peer-student) 3. Weekly presentation: present your weekly results and progress to your lecturer and TA and receive feedback. The chair of the week is in the lead, but everyone participates to the discussion. 4. Weekly report: write your weekly report and receive feedback from your lecturer. The content of the report and adjusted feedback can be easily used for the final report. End Q3: mid-term presentation you will give a presentation in a midterm review. End Q4:: final presentation and report Final symposium: present your results in a 5 min pitch + poster presentation	
Literature and Study Materials	Study material will be provided on BrightSpace.	
Assessment	The final grade is a combination of the following components: Group Grade 1. Mid-term Presentation (10%) 2. Final Presentation (30%) 3. Final Report (30%) Individual Grade 4. Weekly chair Presentation and Report (30%) Please Note: 1. You will need a grade of 6.0 or higher in all components to pass the course 2. Participation/contribution to discussions to SE lectures is expected 3. Participation/contribution to team brainstorming, preparation for the weekly presentation and report is expected 4. Participation/contribution to weekly presentations is compulsory 3. You can receive a bonus/penalty for your team contribution	
Tags	Design Group work Physics Projects	

BM41035	Biomaterials	4
Responsible Instructor	Dr.ir. I. Apachitei	
Instructor	Dr.ir. E.L. Fratila-Apachitei	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In improving patient care, especially in relation with tissue loss or dysfunction, the use of proper biomaterials plays a key role. The shift from tissue removal to tissue replacement and at present, tissue regeneration is driven by the (i)evolution of biomaterials from bioinert to bioactive and bioresorbable and (ii) the increasingly complex biomedical problems of an aging and more active population. As an example, the existing total joint replacements have a 75 - 85 % survivability at 15 years. However, since they were first introduced in the 1960s, average life expectancy increased by at least ten years impacting on the quality of the bone and implants survivability that deteriorate with age. As a consequence, improved implant survivability by 10 - 20 years and alternative treatment methods that delay or eliminate the use of prostheses are badly needed. The Biomaterials course covers in an interdisciplinary approach (biomaterial engineering and biology) the background information on type, composition, processing and properties of biomaterials along with their evolution. Metallic, polymeric and ceramic materials are included and their properties are described and compared with those of biological materials. The main methodologies for biomaterials characterization are covered as well as the procedures for biomaterials standardization and regulation. Further, the interactions between biomaterials and specific biological environments are explained and used for defining the specific requirements and selection principles for biomaterials. Here, biomaterials and medical devices for orthopedics, cardiovascular system and tissue engineering are covered. Finally, the research directions and approaches for development of novel biomaterials are outlined.	
Study Goals	At the end of the course the students will be able to: 1. Identify biomaterials used in: orthopedics, cardiovascular implants, and scaffolds for tissue engineering 2. Define the properties of biomaterials and their property map (range of property values) 3. Describe structure-property relationships for various biomaterials 4. Describe the methodologies for biomaterials characterization and regulation 5. Compare and contrast biomaterials and biological materials 6. Explain the main interactions between a biomaterial and a biological environment 7. Describe the principles for biomaterials selection and design 8. Describe approaches and processes used for functionalization of biomaterials 9. Identify, analyse and describe a biomaterial problem associated with a medical device 10. Explain the clinical performance of biomaterials and medical devices	
Education Method	Lectures, case studies and workshops	
Literature and Study Materials	Lecture slides and recommended literature (reader)	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

BM41050	Applied Experimental Methods: Medical Instruments	4
Responsible Instructor	Dr. J.J. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	In this course, students carry out experimatal reserach in instrument-tissue interactions within the context of application of minimally invasive techniques such as needle or catheter placement. Key in this course is that students familiarize themselves with making an experimental design, setting up an experiment and investigate the response to your experimental manipulations. The results should allow you to critically evaluate the chosen experiment parameters and data analysis methods.	
Study Goals	The student must be able to: Select and apply appropriate research methods. Create an experimental platform. Use suitable methods to analyse the data. Present the results in a concise report.	
Education Method	Students work in groups of two. They acquire basic knowledge through self-study and hands-on experience in experimental work.	
Assessment	Written report (at most 4 pages, two-column scientific paper format).	
Department	3mE Department Biomechanical Engineering	

BM41075	Regenerative Medicine	4
Responsible Instructor	Dr.ir. E.L. Fratila-Apachitei	
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Through regeneration, a living system can restore some of its lost cells, tissue or organs. The fascinating field of regenerative medicine is harnessing the regeneration principles for treatment of human diseases. How to induce and/or control the regeneration process in the adult system is one of the fundamental challenges addressed in this field. After introducing the main concepts and components, the course will present the achievements of regenerative medicine for selected tissues/organs based on a systematic approach including: the pathology involved, the regenerative therapies at molecular, cellular, tissue and/or organ level, and the state-of-the-art results. For each pathology, the following aspects will be discussed: (i) specific requirements for regenerative therapies (e.g., cells type and source, soluble factors, biomaterials/scaffolds), (ii) enabling technologies for cells manipulation, generation and evaluation of scaffolds/tissue/organ, (e.g., organs-on-chip, micro/nanofabrication, bioprinting), (iii) the influence of cells and soluble factors, biomaterials/scaffold properties, and host conditions on the regeneration process, (iv) the relevant cellular and molecular interactions, (v) biocompatibility assessment of regenerative systems and (vi) future challenges for such treatments.	
Study Goals	At the end of the course, the students should be able to: 1. Describe the type of cells, soluble factors, biomaterials and engineering processes used in regenerative medicine 2. Describe and compare various regenerative approaches for a certain pathology 3. Discuss the effects of cell type, biomaterials/scaffold properties and host microenvironment on regeneration processes 4. Identify the main cellular signaling pathways involved in a certain regenerative process 5. Describe models and methods for biocompatibility assessment of a specific regenerative system 6. Propose, justify and describe regenerative approaches/products for a given pathology 7. Recognize the clinical, regulatory and ethical issues of regenerative therapies	
Education Method	Blended education methods including lectures and group projects involving specialists from university medical centers and TU Delft.	
Literature and Study Materials	Lecture slides and selected literature for projects and further reading.	
Assessment	Written exam and project.	
Department	3mE Department Biomechanical Engineering	

BM41090	Computational Mechanics of Tissues and Cells	6
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Instructor	M.J. Mirzaali Mazandarani	
Contact Hours / Week x/x/x/x	0/0/3/3	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	This course is required for students following the Neuromusculoskeletal Biomechanics track of the Biomedical Engineering master programme. Other students are more than welcome to enroll.	
Course Contents	The course consists of two parts: lectures and hands-on workshops. The lectures are designed to give students in-depth knowledge of the mechanical behaviour of cells and tissues and how their behaviour can be explained in terms of mechanical theories. The hands-on workshops give students the opportunity to apply what they have learned in the lectures. A commercial Finite Element (FEM) modelling package is used to make FEM modelling as practical as possible. The emphasis is therefore shifted from the technical details of the FEM modelling towards understanding the modelling procedures and applying them for analysis of certain problems. The problems are chosen to be as close to the real-world problems as possible. In addition to finite element modelling, the principles of medical image processing for finite element modelling are discussed.The course is divided into two almost equal parts. The first part covers patient-specific finite element modelling of tissues and implants. The second part is focused on the mechanical behaviour of cells and proteins (such as collagen). The following topics will be covered in the lecturing (tentative): Lecture 1: Introduction to the course Lecture 2: Geometry of tissues and implants for finite element modelling (including statistical shape and statistical shape and intensity models) Lecture 3-5: Material modelling and material properties (including isotropic and anisotropic linear elasticity, hyperelasticity, viscoelasticity, and poroelasticity) Lecture 6: Loads and boundary conditions (including musculoskeletal models) Lecture 7: Modelling of tissue growth and adaptation Lecture 8: Introduction to cell mechanics, continuum models of the cell Lecture 9: Tensegrity models of the cell Lecture 10: Polymer-based models of the cell Lecture 11-12: Protein mechanics The workshops start from very simple modelling exercises that introduce the FEM package and show how it can be used for simple modelling tasks. The level of difficulty will gradually increase up to the point that students can do FEM modelling of tissues, cells, proteins, and tissue-implant systems.	
Study Goals	Students who have successfully completed this course must be able to: 1.Explain continuum theories that are used for modelling of biological tissues (i.e.anisotropic elasticity, viscoelasticity, and poroelasticity) 2.Translate physiological/pathological conditions of cells and tissues to the language ofmechanics 3.Comfortably use a certain commercial FEM package (possess general FEM modellingskills) 4.Build FEM models of tissues and cells 5.Identify different modelling approaches 6.Make appropriate choices concerning material models, boundary conditions, loadings, etc. 7.Critically assess the results of their FEM models	
Education Method	(Interactive) lectures and hands-on workshops	
Computer Use	Computer will be used for the hands-on workshops	
Literature and Study Materials	Recommended texts for the course: 1.Cowin, Stephen C., Doty, Stephen B., 2007, Tissue Mechanics, Springer. 2.Cowin, Stephan C. (Editor), 2001, Bone Mechanics Handbook 2nd Ed., CRC Press. 3.Bartel, Donald L., Davy, Dwight T., Keaveny, Tony M., 2006, Orthopaedic Biomechanics:Mechanics and Design in Musculoskeletal Systems, Prentice Hal. 4.Mow, van C., Huiskes, Rik, 2004, Basic Orthopaedic Biomechanics & Mechanobiology3rd Ed., Lippincott Williams & Wilkins. 5.Jacobs, Christopher R. et al., 2012, Introduction to Cell Mechanics and Mechanobiology,Garland Science.	
Prerequisites	Students should have working knowledge of the mechanics of materials	
Assessment	Final exam (for the lecturing part) + submission of lecture (individual) and workshop (group) assignments: The grading policy for this course includes: Workshop assignments (total: 60%, first two assignments: 15% each, last workshop assignment: 30%). Lecture assignments (total: 15%, each assignment: 5%) Written exam (25%) 50% of the grade of the written exam is needed to pass the course. At least 2 out of 3 lecture assignments have to be submitted within the deadline provided in the course to pass the course The three workshop assignments are team work activities, and the final report has to be approved by all group members. Once the group members are defined, they cannot be changed. In case of any problems with your group members, we preserve the right to ask for an oral examination. If students fail the re-sit exam, in order to pass the exam in the next academic year, they are asked to not only retake the exam but also submit again two lecture assignments. Any delays with the submission of the lecture assignments will result in down grading of your grade. No delays in the submission of workshop assignment is accepted.	
Department	3mE Department Biomechanical Engineering	

BM41155	3D Printing	4
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Responsible Instructor	Dr. J. Zhou	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	in Track I (Musculoskeletal Biomechanics)and Track II (Medical Devices and Bioelectronics) of MSc Biomedical Engineering BME) and in the specialization of Bio-Inspired Technology (ME-BMD-BITE)	
Expected prior knowledge	BSc in mechanical engineering, materials engineering, industrial design engineering and aerospace engineering with a basic knowledge of materials (e.g. WB2330)	
Parts	- Overview of AM technologies, their principles and applications - Materials selection (polymers, ceramics, metals and composites) - Powder-bed fusion processes - Powdered materials and powder fabrication - Powder characteristics in relation to AM - Structural integration during melting and sintering - Extrusion-based systems - Design for functionality and 3D printability Software issues and AM limitations - Other powder-based manufacturing technologies as alternatives to AM - Post-processing of AM products - Mechanical testing of AM products - Case studies AM for engineering and medical applications - Assignment and presentations	
Course Contents	Additive manufacturing (AM), also known as 3D printing the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to conventional subtractive manufacturing technologies, is set to revolutionize manufacturing across a wide range of industries. The course 3D printing - AM technologies is designed to equip students with the fundamental knowledge of AM and basic hands-on experience in AM. The topics include various AM processes, their principles, capabilities and limitations, materials used for AM processes, powdered material fabrication and characterization, CAD designs for AM, alternative powder-based manufacturing technologies, current and future AM applications, as well as the mechanical evaluation of AM parts. The primary focus is placed on powder-based AM for metallic parts. In addition to a series of lectures, project groups, each composed of 3-5 students, will create their own designs of the objects made of polymer PLA, formulate requirements in terms of geometry, dimensions, tolerances, mass, surface finish, functionality and mechanical properties of these objects, redesign them by making use of SolidWorks, print them, test and evaluate 3D printed products against the requirements, and document the projects in the form of reports and presentations. The new design is expected to reflect the distinctive features and freedom that AM has to offer (thin walls, lattice structures, form-free channels, integrated parts/functions, and customization), but within the capacity of the printing machines (bounding box size, minimum feature size or wall thickness). The report and presentation are expected to contain: reason for the selection of the object, weakness of the existing design, new design features, actual testing results and recommendations.	
Study Goals	Students will be familiar with the concept of AM, various AM processes, AM applications, materials for AM, product design for AM, post-processing techniques and evaluation methods. In addition, students will gain basic hands-on experience in product development via AM. Once successfully completing the course, the students will: - Have a fundamental knowledge of 3D printing/AM technologies for a wide range of applications - Be able to explain the benefits, advantages, challenges and limitations of individual AM technologies as well as alternative powder-based manufacturing technologies - Have an insight into selective laser sintering and melting (SLS/SLM) processes regarding process control in relation to the geometry and mechanical properties of the products - Become familiar with the common (metallic) materials for 3D printing and their characterization techniques - Understand printable geometry design rules and the common methods of digital file creation - Become familiar with the common techniques for post-processing - Become familiar with the common methods for the mechanical evaluation of 3D printed materials and functional parts.	
Education Method	Lectures and student projects Attending lectures and performing assignment are both compulsory	
Computer Use	- Object design with Software (SolidWorks Student Engineering Kit)	
Course Relations	Biomaterials, Tissue biomechanics and other engineering courses in MSc biomedical engineering, materials science and engineering, mechanical engineering, marine technology, aerospace engineering, design for interaction, integrated design and strategic design	
Literature and Study Materials	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Books	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21)	
Reader	- Lecture materials	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	One final exam for 100% of the grade. This can be a written exam or an online exam, depending on the Corona situation. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	M. Loog	

CS4230	Machine Learning 2	5
Responsible Instructor	M. Loog	
Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. D.M.J. Tax	
Instructor	Dr.ir. J.H. Krijthe	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	This course is the more advanced and research oriented follow-up to CS4220 [Machine Learning 1]. The content of the latter is, therefore, expected as prior knowledge.	
Course Contents	The course will treat a number of machine learning theories and techniques in detail and on an advanced level. In part, the contents of the course revolves around the research focus and interests of the lecturers teaching in the course. Possible topics : learning theory, latent variable models, online learning, Markov decision processes and abstraction, semi-supervised learning abstraction, active learning, learning curves, causal reasoning and discovery...	
Study Goals	After successfully completing the course, the student is able to apply the techniques and theories that have been covered in the course. In addition, they are able to develop learning strategies for new and previously unseen situations. Moreover, the student can provide reasoned justifications for these strategies based, for instance, on theory and/or experiment.	
Education Method	Lectures + Q&A sessions	
Assessment	Grading is based on two parts. Following the lectures -- we have about 11 of those, there is an individual assignment that will be graded pass/fail. In addition, there is a written examination that will be graded on a scale from 1 to 10. You pass the course when you both have a pass for the assignment and a passing grade for the written exam. Upon passing the course, your final grade will be the grade for the exam. Finally, note that there are two possibilities offered for the written examination, but not for the report.	

CS4255	Algorithms for sequence-based Bioinformatics	5
Responsible Instructor	T.E.P.M.F. Abeel	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Bioinformatics analyses in genomics aim to compare large sets of genomes in order to understand and explain differences in traits of an organism. Contemporary methods are powered by fundamental algorithms and data structures, which are efficient and scale to large data sets. A thorough understanding of these algorithms and data structures is necessary for advanced users and developers in this area. In addition, understanding how comparative genomics is developing is important to shape your own research. In this course, we will cover genome analysis, variant analysis, and pangenomics. Core concepts, applications, and future trends will be discussed, with a focus on the algorithms and data structures underlying state-of-the-art methods.	
Study Goals	After having followed this course, the student has a good understanding of algorithms and data structures in genomics used for DNA sequence analysis. The student is able to implement algorithms in python, and can translate methods described in scientific literature into a working implementation.	
Education Method	The course is offered as a mix of lectures, exercises and a project	
Books	http://bioinformaticsalgorithms.com	
Assessment	60% graded exercises, 40% project report. A resit opportunity is available for both components Students must receive a 'pass' mark for each component individually (>=5.8). Notes: - No scores and/or submissions are transferred to the next year or from previous years. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CS4329	Recent topics in bioinformatics	5
Responsible Instructor	Prof.dr.ir. M.J.T. Reinders	
Instructor	Dr. J.S. de Pinho Gonçalves	
Instructor	T.E.P.M.F. Abeel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Bioinformatics is at the heart of many modern systems biology analyses, and encompasses the application of statistics and computer science to (large-scale) biomolecular datasets. In essence, bioinformatics is about smart ways of extracting knowledge from the enormous amounts of data that can be generated using modern measurement techniques. For instance, it plays an important role in finding the genetic origins of various diseases, such as cancer, diabetes or alzheimer. Covid-19 disclaimer: all information here is correct at the time of writing. This may change based on developments around COVID-19. Students will be informed as soon as possible through Brightspace.	
Study Goals	After successfully completing this course, the student is able to: explain several high-throughput data acquisition experiments, such as DNA/RNA sequencing, and discuss the benefits and limitations of these methods comprehend the statistical and computer science issues in analyzing high-throughput data discuss the basic systems biology approach, and the role of high-throughput measurements, gene selection and classification therein explain bioinformatics methods, algorithms and models to a non-expert audience implement or execute basic algorithms from descriptions provided in scientific literature read and comprehend a current scientific paper and reflect on the bioinformatics methods used in such paper	
Education Method	In this course we will study some key examples of bioinformatics analyses by reading a set of selected papers that present some significant biological conclusions.The course is run a flipped class-room students take turns guiding us through the selected papers. Teachers moderate the discussion and fill in any gaps in understanding.All students are required to read, and prepare the course material before every lecture to effectively participate. Each week there are one or two lectures, each of 90 minutes In each lecture one paper (the course material) will be discussed in detail. One or more students will present and explain the details of this paper. It is essential that you highlight the (bioinformatics) methodology of the paper. The schedule for this will be prepared in the first lecture. This presentation is graded. All students are expected to have read the paper and should have an active role in the discussion about the paper. Each student should be prepared to raise at least two critical remarks or questions during the discussion. The quantity and quality of participation of each student is graded. The student will perform a practical project in groups where they gain practical experience in state-of-the-art bioinformatics methods.	
Assessment	- Presentation (20%) - Participation during discussions (20%) - Reporting about unseen paper with oral presentation (60%) disclaimer: information may change depending on the developments around the coronavirus.	

EE4650	Advanced Magnetic Resonance Imaging	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	A. Webb	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Vector algebra and vector calculus, linear circuit theory, signals and systems (Fourier transform, steady-state analysis), linear algebra and differential equations, electromagnetics (BSc level or higher)	
Course Contents	General introduction, history of MRI, general description of an MR scanner, general principles of MRI, applications and current research themes. General and brief description of spin, restriction to particles with spin ½ (proton), equilibrium magnetization for spin ½ particles (proton), the Bloch equation, general description of the gyromagnetic ratio, T1 and T2 relaxation times. Electromagnetic fields: Maxwells equations, the electromagnetic boundary conditions, the temporal and spatial Fourier transform, time-harmonic fields, the polarization state of an electromagnetic field (linear, circular, elliptical), Poyntings theorem (power balance), and reciprocity. Flipping the magnetization: received signal in terms of the electromagnetic receive field. Solutions of Blochs equation for various magnetic fields, the most important formula in MRI: formula for the Larmor frequency, chemical shift, scalar coupling, basics of 1-DNMR spectroscopy, elementary measurement sequences, spin-echo, inversion recovery, measuring T1 and T2, introducing T2*. Using gradients to spatially encode signals, k-space formalism, image reconstruction via multi-dimensional Fourier transforms, simple gradient echo and spin echo pulse sequences, three-dimensional imaging, angiography. Background knowledge for hardware design: Maxwells equations and quasi-static fields, the law of Biot-Savart, spherical harmonics. Interaction of RF with tissue as a function of frequency, different media/tissue types and the constitutive relations. RF coil and array design: design of RF coils and RF arrays, the Specific Absorption Rate (SAR). The electromagnetic field due to an electric dipole (small current-carrying piece of wire) and a magnetic dipole (small current-carrying loop). Gradient coil design and the static background field: Realizing gradient fields, Helmholtz and Maxwell coils, advanced gradient coil design using stream functions, consequences of nonuniform static background fields. Numerical modeling, software packages, body models, coil models. Measurements and imaging in practice, SNR and other practical aspects. Current/future research topics, parallel imaging, sparse sampling, electrical properties tomography	
Study Goals	After completing this course, the students will: 1. Understand the principles of magnetic resonance from both a quantum and classical mechanics point-of-view, 2. Understand the interaction of strong electromagnetic fields at high, medium and low frequencies, how these can be manipulated, and the safety features associated with these fields, 3. Understand how spatial information is encoded into magnetic resonance images, the origins of image artifacts, and the signal processing algorithms involved in image reconstruction, 4. Understand the relaxation phenomena associated with magnetic resonance, and how these are important in determining image contrast, 5. Understand how the different components of magnetic resonance systems are designed using Maxwells equations, spherical harmonics, and the Biot-Savart law, 6. Understand the current state-of-the-art of MRI hardware and image processing, and the new opportunities for future developments.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes. Will be made available via Brightspace.	
Reader	Lecture notes. Will be made available via Brightspace.	
Assessment	Assignment and oral exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IFEEMCS4250	Statistical Learning for Engineers	4
Responsible Instructor	Dr. H.N. Kekkonen	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra, Calculus, Probability& Statistics on the level of TU Delft Bachelor; some programming skills in e.g. Python or R.	
Course Contents	Linear regression, classification, linear model selection and regularisation, tree based methods, artificial neural networks, and deep learning.	
Study Goals	After this course you should be familiar with several commonly used statistical and machine learning methods. know how to estimate unknown parameters and make predictions based on data. be able to apply statistical learning techniques using Python or R. know how to evaluate model accuracy and be aware of the strengths and limitations of different models.	
Education Method	Lectures and exercise + computer labs	
Assessment	Final exam and homework assignments.	
Enrolment / Application	Please note: students do not need to register via Osiris for this course, only for the corresponding exams. Students only have to enroll via Brightspace.	
Co-Instructor	Dr.ir. W.T. van Horssen	
Co-Instructor	Dr. N. Parolya	

IN4089	Data Visualization	5
Responsible Instructor	T. Höllt	
Instructor	Prof.dr. E. Eisemann	
Contact Hours / Week x/x/x/x	0/2/0/0 & lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Master course MKE/ST/DS	
Expected prior knowledge	IN2905-A/IN4152/CSE2215 Computer Graphics (recommended, not required). The practicals will be implemented in HTML/Javascript/D3 (InfoVis) and C++ (VolVis). We consider programming skills as a requirement but not necessarily in the mentioned languages and no advanced skills are needed. Relevant topics will be introduced and experience in other programming languages should make adaption feasible.	
Course Contents	Data visualization is the visual representation of data by computer generated images. The data sets can be results of numerical simulations or measurements (scientific visualization), or other data collections such as databases (information visualization). The goal is to improve insight, understanding and/or communication of data. Data visualizations use a combination of methods from a very diverse variety of disciplines: perception, computer graphics, human computer interaction, algorithmics, image processing, machine learning, numerical analysis, optimization, The course has two main parts; information and scientific visualization that will involve knowledge of diverse disciplines. As a computer science course, affinity to algorithmic thinking and programing skills will be needed. Topics covered: models of the visualization process; colour models and use of colour; information visualization; representation and processing of data; volume visualization; interactive visual data analysis; visualization of vector fields and flows. Guest lectures might be given on selected topics.	
Study Goals	The goal of the course is to get knowledge on the fundamentals that are part of data visualization. The main principles and techniques that are the basis of generating effective visual representations of data. Techniques and cases of data visualization are discussed. There are several applications for the techniques, such as medical, engineering, finances, economics, game analytics, and more. By the end of the course, you should be able to LO1: Discuss a large range of visualization techniques. LO2: Discuss a perception principle of visualization. LO3: Explain mathematical principles and algorithms of visualization techniques. LO4: Design suitable visualization systems for a given practical data analysis problem. LO5: Implement visualization systems for a given practical data analysis problem.	
Education Method	Lectures, practical assignments, self-study, and projects.	
Literature and Study Materials	Course slides, instructions for projects, and selected literature. Chapters from: Visualization Analysis and Design Author: Tamara Munzner CRC Press Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann All available in electronic form via Brightspace or at TUDelft library.	
Assessment	The final grade is a weighted average based on two visualization projects, and a written exam that might contain multiple choice questions. The projects will be developed in groups of 1-3 and are evaluated based on the reasoning/justification of the techniques used based on the material given at the course, effectiveness of the results, technical contribution or implementation, quality of the documentation and presentation. disclaimer: information may change depending on the developments around the coronavirus.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Judgement	The grade consists of 3 elements: Information Visualization project, Volume Visualization Project and a written exam. The two projects will be developed in couples and will represent 70% of the mark together. All projects, which are handed in late will be evaluated with a zero and impact the part of the mark that corresponds to the project. Additionally, a written exam will be held, which will represent 30% of the mark. The exam might contain multiple-choice questions. The project is evaluated based on the developed result, its documentation and presentation. Final Mark = 0.35 InfoVis Project + 0.35 VolVis Project + 0.3 Exam Regarding minimal grades for partial examinations: A passing final grade for a course can only be earned when for all component examinations and practicals of that course at least a 5,0 is earned, and the average grade for all components is at least a 5,8 (Article 17 RRBE (subsection 6)) The exam will have a resit. No resit will be provided for the projects unless the mark on the exam (NOT the resit exam) and the other project are above 7.5 The project resit is not automatic and must be initiated by you within two weeks of the grades being published. It will be evaluated at individual bases, despite the project being done in groups. Resit of a project will mean starting a new project.	

LM3311	Green Chemistry and Sustainable Technology	3
Responsible Instructor	Dr. F. Hollmann	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4 5	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Organic Chemistry (LB1045 or 4051OCSTRY) Chemical Biology (LB2621) OR Catalysis (4052KATALY)	
Course Contents	Depleting fossil fuels and environmental concerns force us to change the way we produce our goods. The transition of the chemical industry to becoming sustainable represents one of the major challenges to be faced in the 21st century. Catalytic reactions will increasingly substitute the existing energy- and resource-intensive production procedures; hence reducing or preventing toxic pollutants. Biotechnology will play a major role in this transition and complement advanced chemical production technologies. This course will highlight the emerging role of catalysis in the chemical industry. A major focus will lie on biotechnological production of chemicals (fermentation processes, isolated enzymes). These methods will be related to their chemocatalytic counterparts. Thus, ample examples including the underlying mechanistic principles - of bio- and chemocatalytic processes will be discussed. Finally, the emerging field of renewable feedstocks will be discussed not only from a chemical standpoint, but also on socioeconomical and ecological grounds. The 12 Principles of Green Chemistry will serve as guideline to assess the scope and limitations of the examples discussed.	
Study Goals	After attending this course the student will: (1) have detailed knowledge of sustainable catalysts (enzyme- and chemo-catalysis) (2) know industrial examples of sustainable manufacturing practice (3) be able to use the 12 Principles of Green Chemistry to assess the sustainability of a given process (4) be able to critically assess the purport of green claims	
Education Method	Lecture Sheets, specialized books, articles and patents	
Literature and Study Materials	selected articles will be available via Brightspace	
Books	suggested reading: 'Green Chemistry - An Introductory Text' (2nd ed., M. Lancaster) RSC Publishing (ISBN: 978-1-84755-873-2) and/or 'Green Chemistry and Catalysis' (Green Chemistry and Catalysis R. A. Sheldon, Isabella Arends, Ulf Hanefeld), Wiley VCH (SBN: 978-3-527-30715-9)	
Assessment	(1) Written exam (digital off-campus (open book) or on-campus, depending on the COVID-19 measures), 75% of the overall grade (2a) only for SET students: written assignment about a topic of Green Chemistry (groups of 3-4 students), 25% of the overall grade (2b) for non-SET students: group presentation about a topic of Green Chemistry (groups of 4-5 students), 25% of the overall grade	

LM3432	Analysis of Metabolic Networks	6
Responsible Instructor	Dr. W.M. van Gulik	
Course Coordinator	Dr. W.M. van Gulik	
Instructor	T.W. Páez Watson	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The focus of AMN is on the quantitative analysis and understanding of living cells and their interaction with the environment. Especially, specific rates (growth, production) which result from genetic properties and a highly connected set of networks: - Metabolic reaction networks - Genetic regulation networks During the course we will discuss different modelling approaches to evaluate and predict cellular metabolism. This includes prediction of the effect of genetic and environmental changes on rates in biological systems. Modeling is useful to test hypothesis and understand observed behavior. For a successful modeling cycle, the connection with the experiments is essential - the course addresses experimental methods to quantify fluxes in networks We will present three levels of detail in the course: - Black box kinetics & thermodynamics (refreshment of Bachelor courses and some new) - Medium scale stoichiometric models - Small scale kinetic models Further the course contains: Refreshment of prior knowledge (mass balances, conversion rates, black box kinetics/ stoichiometry, energy-, product-, growth pathways) Validation of experimental data & data reconciliation Metabolic flux analysis of large networks Transport over cellular membranes Kinetics of metabolic pathways Mathematical modeling of central carbon metabolism Regulation of enzyme expression, principles & modeling Sources of bio-information/ databases To perform the calculations, python is used as simulation tool for biological processes.	
Study Goals	After the course the student should be able to - Set-up massbalances for bioreactor processes - Use linear algebra to estimate and reconcile metabolic rates - Relate metabolic network structure to maximal biomass and product yields/ product rates - Show understanding of the thermodynamic and redox constraints in metabolic networks - Define experiments to calculate fluxes - Set-up mathematical models which relate network/ enzyme properties to fluxes - Interprete results from stoichiometric and kinetic modeling to define engineering targets. Most relevant transferable skills - Communication - Problem solving - Teamwork - Time management	
Education Method	Lectures and exercise classes	
Assessment	Group assignments (50%), transferable skills acquired are teamwork, communication, time management, problem solving Final individual exam (50%)	
Special Information	For the 3rd course of their specialization Biochemical Engineering, students can make a choice out of LM3611 Microbial Community Engineering LM3432 Analysis of Metabolic Networks	

LM3442	Metabolic Reprogramming	6
Responsible Instructor	Prof.dr.ir. P.A.S. Daran-Lapujade	
Instructor	Prof.dr.ir. P.A.S. Daran-Lapujade	
Contact Hours / Week x/x/x/x	0/3/4-8/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Required for	MSc specialisation Cell Factory, Life Science & Technology	
Expected prior knowledge	This course requires BSc-level knowledge in Microbial Physiology, Molecular Genetics and insight into the quantitative aspects (yields, rates) of Biotechnological processes (LST BSc courses LB2762TU, LB2161 and LB1511TU or equivalents thereof).	
Parts	Lectures schedule: 1. General introduction to metabolic reprogramming and the course First lecture on increasing fluxes and yields in existing pathways 2. Classical strain improvement and reverse engineering 3. Evolutionary and reverse engineering 4. Synthetic biology, new products 5. Heterologous protein production Journey into the publication process 6. Mid-term exam and collegial correction	
Course Contents	Application of targeted genetic modification and rational application of evolutionary strategies for the optimization of existing product formation pathways in micro-organisms, introduction of new product formation pathways, physiological constraints on process optimization	
Study Goals	After this course, the students should: - understand fundamental aspects, possibilities, limitations and challenges in the rational optimization of microbial cells for industrial product formation - be able to critically read scientific papers and discuss their contents with colleagues in a structured manner	
Education Method	This course is a combination of a lecture series and presentations/discussions of scientific publications. The lecture series will consist of 6 three-hour sessions and will introduce key principles of metabolic reprogramming and introduce the format of the second part of the course. For this second part, a collection of relevant scientific publications will be handed out. Each student participating in the course prepares a critical (oral) presentation on one of these papers. Furthermore, all students read the complete set of publications, which allows them to participate in plenary discussions (led by the teachers) on the papers. In addition to the scientific content, attention will be given to key quality aspects of the scientific publications (style, statistics, internal consistency, impact). Sessions of 2 hours will be used for the presentation and discussion of two scientific papers. Education and exams will be on site.	
Literature and Study Materials	A set of scientific publications will be provided by the teachers at the outset of the course (via Brightspace)	
Assessment	Students will be assessed based on: 1. A short mid-term exam which goal is to assess student's ability to critically analyse scientific literature. (10% of final grade) 2. Their presentation and critical analysis of a chosen scientific publication. (25% of final grade) 3. Their critical analysis of a chosen article and scientific discussion with the group (called Critical Panel, 20%) 4. Their contribution to the discussion following presentations by their fellow participants. (20% of final grade) 5. A final, written exam that will cover the subject matter taught in the lecture series as well as key elements discussed during the evaluation of the scientific paper. (25% of final grade) While students are encouraged to analyse together scientific literature, assessment of the course is purely based on individual performance. Resits in period 4 by written or oral examination.	
Permitted Materials during Tests	Pen, paper, calculator, annotated article (supplied in advance)	
Enrolment / Application	not later than one week before the first lecture via BrightSpace	
Location	Department of Biotechnology, Delft University of Technology	

LM3451	Bioprocess Integration	5
Responsible Instructor	Dr.ir. A.J.J. Straathof	
Module Manager	Dr.ir. A.J.J. Straathof	
Contact Hours / Week x/x/x/x	4-6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This is an introductory, mandatory course for the MSc LST and prepares especially for the Design project, for Fermentation Technology & Environmental Biotechnology, and for Transport & Separation	
Expected prior knowledge	Basic biosciences, process technology, bioprocess design, chemistry and mathematics as required for admittance to the MSc programme Life Science & Technology. The expected biochemical engineering background is: P.M. Doran, Biochemical Engineering Principles, Academic Press 1995, chapters 1-9 and 11, except pages 139-163. Or the equivalent in the 2nd edition of this book. The expected bioprocess design background involve capability - to study and use industrial biotechnology literature - to calculate fermentation stoichiometry - to formulate simple bioprocess block schemes - to calculate mass flows in such processes and assess associated economic/ecologic/social impact	
Course Contents	- Short recap of bioprocesses in industrial and environmental biotechnology - Stages in the bioprocess design cycle that are treated in this course: design definition, synthesis, analysis, process assessment - Flow sheeting and process simulation using dedicated software (SuperPro Designer), to solve mass and energy balances of bioprocesses, to size equipment, and to perform economic assessment - Operation modes: batch/continuous; cocurrent/countercurrent/crosscurrent - Bioprocess unit operations and their short-cut design: enzymatic conversion, fermentation, sedimentation, centrifugation, filtration, membrane permeation, L/L extraction, adsorption, ion-exchange, chromatography, evaporation, distillation, precipitation,crystallization - Selecting and sequencing unit operations; integration of bioprocess functions; in-process recycles; waste destinations - Batch scheduling and debottlenecking - Energy streams in bioprocesses; heat integration - Economic assessment method of SuperPro Designer; sensitivity of process performance to microbial and enzymatic parameters and process settings	
Study Goals	After the course the student should be able to - describe the functions and the functioning of the different parts of biotechnology processes - draw block schemes of bioprocesses suitable for detailed analysis - use software for bioprocess simulation in order to obtain mass and energy balances, equipment sizing, batch scheduling, heat integration, and economic assessment - perform off-line short-cut calculations on sizing the most relevant unit operations in biotechnology, on batch scheduling, on heat integration, and on economic assessment - identify semi-quantitatively the potential impact of protein or metabolic engineering on overall improvements of a production process	
Education Method	- Lectures - Tutorial - Training exercises in classroom - Assignments	
Computer Use	SuperPro Designer (will be instructed)	
Literature and Study Materials	- Lecture slides handouts - Supporting documents - Training exercises + solutions - Old exams + solutions All will be posted on Brightspace during the course.	
Assessment	During the course three assignments on a case study in teams of up to three students (unrounded assignment grade counting for 50% of the unrounded overall grade) and a final individual written exam (unrounded exam grade also counting for 50%). But if the rounded overall grade is >5.5 while the rounded grade for the exam is <5, the rounded overall grade will be decreased to 5.5. Calculation examples: - Assignments 4.70, exam 9.30, rounded overall grade 7.0 - Assignments 9.30, exam 4.70, rounded overall grade 5.5 The exam has a resit. Then the assignments grade will be kept for recalculating the overall grade. Assignments can be redone only in a next year. Then, the exam grade will be kept, unless the exam is also redone.	
Enrolment / Application	Before the first lecture via Brightspace	
Studyload/Week	4-6 contact hours per week; furthermore, about 8 hours/week for self study and assignments. At the end, a full week for the exam incl. preparation	
Continuing Courses	Design project LM3822	
Schedule	Contact hours that involve lecturing an exercising: Week 1.1: Introduction (1h), Bioprocess simulation training (5h) Week 1.2: Biokinetics and bioreactors (4h) Week 1.3: Batch scheduling (2h); Solid/liquid separation (2h) Deadline assignment 1 Week 1.4: Contacting and extraction (4h) Week 1.5: Evaporation/drying/distillation (2h); Adsorption/ion-exchange/chromatography (2h) Deadline assignment 2 Week 1.6: Precpitation/crystallization; Membrane operations; Design definition; Process assessment (each 1h) Week 1.7: Bioprocess energy (2h); Bioprocess synthesis (2h) Deadline assignment 3 Some swapping may occur between weeks 1.5 and 1.6 depending on the needs of assignment 2. Weeks 1.3, 1.5, and 1.7 include also 2h support for assignments.	

A detailed lecture schedule will be published on Brightspace before the course starts.

LM3561	Ethical, Legal and Social Issues in Biotechnology	3
Responsible Instructor	Drs. L. Asveld	
Instructor	Drs. L. Asveld	
Contact Hours / Week x/x/x/x	0/0/0/8 hrs/week	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	This course is open to MSc students with a relevant background (LST, Chemical Engineering, nanobio).	
Course Contents	In this course you will explore ethical, legal and societal issues of biotechnology that will be relevant to your future work as a professional in biotechnology. The course will introduce concepts in ethics in the life sciences and familiarize you with some of the most prominent ethical theories and their applications. The course will also stimulate you to critically reflect on the design assignment you do parallel to this course and to integrate a wide range of societal considerations into your design assignment. This course is a combination of lectures and seminar sessions, coupled with a reflection on your design project and other topics of interest.	
Study Goals	After having completed the course you: will be able to recognise and analyse ethical, legal and societal issues related to biotechnology. will be able to apply methods to create societally robust and ethically acceptable designs	
Education Method	The course is highly interactive, combining lecture and seminar discussions with student presentations.	
Literature and Study Materials	Selected texts and other online material available on Brightspace	
Assessment	50 % group assignment related to your design project (1 presentation) and 50 % individual written assignments (1 essay).	
Enrolment / Application	Enrolment via Brightspace (courses > Master > LS&T) is required one week before the first lecture.	
Schedule	Q4	
Location	Delft	

LM3581NB	Metabolic Systems Biology	3
Responsible Instructor	Dr. G.E. Bokinsky	
Course Coordinator	Dr. G.E. Bokinsky	
Contact Hours / Week x/x/x/x	0/0/2-3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Systems biology is an integrated approach of experimental and mathematical methods that aims a comprehensive understanding of how biological functions arise and how they are regulated. We will focus on different levels of cellular regulation in bacteria: metabolism, gene expression, and global physiology, while emphasizing interconnections between the regulatory levels to illustrate how stable cellular phenotypes occur, and how these interconnections give rise to seemingly-miraculous properties, like growth rate maximization. The course will consist of a mixture of lectures and student presentations and discussions of current literature. Experimental and modelling approaches will be discussed. For unravelling how cell functions are regulated, different platforms have been developed, among which are techniques named -omics technologies, e.g. metabolomics, transcriptomics, genomics and proteomics. However, data alone are insufficient for understanding how a metabolic pathway or a signalling cascade gives rise to different phenotypes. Using literature presentations and discussion, this course introduces the uses measurement technologies for the purpose of: - Quantifying metabolites within bacteria - Quantifying genetic regulation in bacteria - Quantifying single-cell gene expression - Identification of interactions between genes and metabolism The course presents: - Measurement techniques and data interpretation for protein and transcript levels at both ensemble and single-cell level - Principles and functions of gene expression networks, emphasizing autoregulation and regulatory dynamics - Using modelling at different levels of detail to explain cellular behaviors and adaptive advantages - Using modelling to identify sites of regulation in metabolic pathways Note that online or in-person classroom attendance is mandatory at all sessions.	
Study Goals	After this course the students should be able to: - Understand how models published in system biology journals are used to explain experimental results - Predict dynamics of cellular networks in response to perturbations - Explain how regulatory connections and dynamics result in a defined phenotype - Explain how data and models are used to investigate cellular behavior - Critically evaluate literature papers in field of systems biology - Communicate findings from literature (and criticism of the literature) to an audience of non-experts	
Education Method	Lectures Peer lectures Computer practicals Classroom Discussions	
Assessment	- Computer Exercise on gene transcription (25 %) - Literature case study: Oral presentation of one journal article and leading a classroom discussion, either alone or as part of a group (50%) - Participation in classroom discussion (15%) - Short written analyses of assigned reading (1 paper per week) (10%) There will be no final exam.	

LM3601	Molecular Biotechnology & Genomics	6
Responsible Instructor	Prof.dr.ir. J.G. Daran	
Instructor	Prof.dr.ir. J.G. Daran	
Contact Hours / Week x/x/x/x	0/3/4/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Course Contents	The Molecular Biotechnology and genomics course offers an in-depth overview of contemporary biotechnology. The program aims at providing the student with a profound fundamental basis, as well as offering him an integrated study package, compiled from a wide scope of scientific disciplines, which together underscore the practical realizations of application-oriented molecular biotechnology.	
Course Contents Continuation	1. Genomes (strains, genome sequences, sequencing technologies,). Origin of industrial strains/applications Sequencing industrial organism, why? Methods from Sanger to next generation sequencing techniques Analysis of genome sequences 2- Strain improvement strategies. Classical strain improvement High throughput screening genome shuffling Molecular Genetics/Metabolic engineering Examples 3- Molecular genetics (strain design) Genetic engineering solutions Deletion, silencing, overexpression Vector design Restriction versus recombination) Host design integration HR vs NHEJ 4 Cloning strategies Identification of industrial enzymes by expression cloning 5-Transcriptomics. 6- Computer classes Plasmid design reference guided and de novo geneme assembly	
Study Goals	After this course the students should: Obtain a deeper understanding of the possibilities and limitations of molecular cell biology/molecular genetics and genomics applications in modern biotechnology. Integrate knowledge on (i) molecular cell biology/molecular genetics, (ii) genomics and (iii) cell factory research.	
Education Method	E.G.: Lecture series and hands-on exercises Computer classes to acquire hands on experience. This course is a combination of a lecture series and in-silico practical exercises The course is divided in two parts: in the first a lecture series accompanied with relevant exercises will introduce key concept in molecular genetics. The second part will be based on do it yourself genomics classes aiming at teaching principles of genome data, assembly analysis. The computer sessions will be punctuated with explanatory lectures.	
Assessment	Students are evaluated by 3 different methods. 1. Two individual assignments are proposed to the students during the course, two in the first period and one in the second. The students have between 2 to 3 weeks to complete it and to give it back to teacher before a clearly defined deadline. Each assignment accounts for 20% of the final grade. 2. Time limited (3h30) assignment on genome assembly. Due the complexity of the exercise, this assignment needs to be tightly supervised to allow the students to progress. The grade of this assignment is used as a bonus (up to 10% of the final written exam grade) added to the grade of the final written exam. 3. Written exam (60% of final grade of the final grade). All grades are individual	
Enrolment / Application	not later than one week before the first lecture via Brightspace Delft (courses > Master > LS&T)	
Location	To be mentioned on due course. See also: www.timetables.tudelft.nl	

LM3611	Microbial Community Engineering	6
Responsible Instructor	Prof.dr.ir. M.C.M. van Loosdrecht	
Responsible Instructor	D.G. Weissbrodt	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc-level knowledge in Microbial Physiology (LST BSc-course LB2762TU or equivalent thereof)	
Course Contents	The course will discuss the metabolic potential of microbes and their role in the cycling of chemical elements. It will indicate the fundamental aspects of microbial competetion and cooperation in natural and man-made environments. Specific attention is given to the translation of ecological insight in the development of microbial community-based bioprocesses, such as wastewater treatment, biogeochemistry, biocorrosion, and the production of chemicals and bioenergy.	
Study Goals	After this course, the students should be able to: - Describe the concept and methodology of natural selection-based microbial community engineering as a means for the development of bioprocesses. - Capture the impressive microbial diversity in nature and its potential application in biotechnology. - Process the global element cycles and the organisms that catalyze the reaction in these cycles. - Characterize and implement the different diagnostic tools that are currently available to study the diversity and activity of microorganisms in natural and man-made environments. - Formulate scientific research questions to investigate natural and technical microbiomes, develop an investigation method, and predict the distributed functional performance of the microbiome. - Write a concise scientific essay / perspective article on a selected topic in microbiome science & engineering.	
Education Method	Microbime science and enineering is a very indterdisciplinary field. The course is built on a lecture series, given by specialists of their fields. Each lecture will bring the basics and broadens on the applicability of the concepts. Besides personal development on concepts and methods, peer learning as well as critical and creative thinking will be fostered via the production of an essay / perspective article. The highlights of the essay will be presented in a poster presentation. The MCE learning achievements will be assessed on an individual written exam. T	
Literature and Study Materials	- Session handouts on Brightspace - Chapter: Weissbrodt, Laureni, van Loosdrecht, Comeau (2020) Basic Microbiology & Metabolism; in: Biological Wastewater Treatment: Principles, Modeling and Design. 2nd Edition, IWA Publishing, UK. - Optional book: Madigan, Bender, Buckley, Sattley, Stahl (2018) Brock Biology of Microorganisms, 15th Edition, Pearson Higher Edudaction, USA.	
Assessment	Learning objectives: Fundamentals of MCE Core cycles in MCE Advanced methods in MCE Concepts integration Critical thinking and scientific creativity Assessment methods: (1) Individual written examination (50% of course score): ONSITE or ONLINE EXAMINATION depending on COVID-19 situation and in-house regulations. (2) Group essay (50% of course score) >> 2.1 Group manuscript (30% of course score) >> 2.2 Group presentation (10% of course score) >> 2.3 Individual assessment in presentation (10% of course score)	
Enrolment / Application	Not later than one week before the first lecture through Brightspace (courses > Master > LS&T)	
Special Information	For the 3rd course of their specialization Biochemical Engineering, students can make a choice out of LM3611 Microbial Community Engineering, or LM3432 Analysis of Metabolic Networks	
Remarks		
Schedule	(the tentative schedule can be subjected to changes) Part 0. Introduction to Microbial Community Engineering (MCE) > MCE Introduction / Weissbrodt Part 1. Fundamentals of MCE > Microbial ecology principles / Weissbrodt > Redox reactions in microbial conversions and growth / Kleerebezem > Thermodynamics of microbial growth / Kleerebezem > Role of biopolymers in microbial ecophysiology / van Loosdrecht > Microbial competition along gradients and biofilms / Picioreanu Part 2. Core Cycles and Conversions in MCE > Nitrogen bioconversion network / Laureni > Metabolic puzzles on the nitrogen cycle / Kleerebezem > Organic carbon cycle: anaerobic conversions and chain elongation / Kleerebezem > Phosphorus cycle and granular sludge / van Loosdrecht > Extracellular polymeric substances / Lin > Bioelectrochemical conversions / Jourdin > Sulfur cycle / Sorokin > Phototrophic conversions / Weissbrodt + Cerruti Part 3. Classical and Advanced Analytical Methods for MCE	

	> Community systems microbiology and ecophysiology / Weissbrodt > Metagenomics and metatranscriptomics / Weissbrodt > Introduction to proteomics / Pabst > Metaproteomics applications / Pabst > Metabolic flux analysis in mixed-culture systems / Wahl > Resource allocation to predict microbial performance / Wahl Part 4: Concepts and Data Integration in MCE > Concepts and data integration in MCE / Weissbrodt > Numerical methods for MCE / Weissbrodt + Calderón Franco > MCE Conclusion / Weissbrodt EBT research projects and lab tour / Laureni + PhDs + postdocs
Location	TU Delft and/or on-line depending on public health condition and regulations

LM3701	**Advanced Enzymology**	**6**
Responsible Instructor	Dr.ir. P.L. Hagedoorn	
Instructor	Dr.ir. P.L. Hagedoorn	
Instructor	Dr. D.G.G. McMillan	
Contact Hours / Week x/x/x/x	0/4/3/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc LST, BSc MST or equivalent in terms of Biochemistry/Chemistry	
Course Contents	The course Advanced Enzymology covers basic biochemical principles, which are applied to understanding catalysis and energy conversion by (metallo-)enzymes, their biological function and their dysfunction in disease. This course also focuses on theoretical and practical backgrounds of various spectroscopic, electrochemical and pre-steady state kinetic methods, as well as on the use of computer visualization programs and the enzymology of membrane enzymes. Students will learn how to employ these methodologies to the determination of the structure of metallo-centers and function in terms of the molecular mechanism of catalysis. To successfully apply metalloenzymes in research or industrial applications, specific problems related to the (over)expression of metallo-proteins such as cofactor biosynthesis - are addressed. All these aspects will be illustrated by using two important enzymes: cytochrome P450 and methane mono-oxygenase.	
Study Goals	After the course the students should - be able to apply basic bio-inorganic principles to metallo-enzyme catalysis, understand the biological role of metallo-enzymes and the consequences of their dysfunction in disease - have obtained a basic theoretical level in various spectroscopic and kinetic methods, in electron transfer theory and be able to apply theory to the determination of metallo-enzyme structure and molecular mechanism - be able to judge the scope and limitations of metallo-enzymes in industrial applications - be able to design (metallo)enzyme production, isolation and assay procedures and visualize 3D enzyme structures - compose a strategy for the design of a novel biocatalyst for a desired chemical reaction - use theoretical knowledge of lipid membranes to understand the mechanism of membrane transport proteins and membrane enzymes - show responsibility to complete an individual assignment on time with sufficient effort - able to independently solve a scientific problem related to advanced enzymology	
Education Method	The course is given as a series of live (online or on-campus) lectures of 3 hours (6 times) and 4 hours (6 times). After each lecture a formative assignment will be handed out as a self-assessment assignment on Brightspace, which will be discussed during the subsequent lecture. The course includes a 4 hour computer practical on enzyme structure visualization and analysis using PyMOL software. During the course the students will also read recent scientific papers and discuss the contents.	
Literature and Study Materials	Syllabus and lecture slides	
Assessment	Written exam (digital off-campus or on-campus, open book; 2/3 of final grade) and an individual assignment (1/3 of final grade). The following transferable skills will be included in the assessment of the individual assignment: planning and problem solving (see study goals).	
Enrolment / Application	not later than one week before the first lecture via Brightspace Delft (courses > Master > LS&T)	
Schedule	Metals in biology - Introduction bio-inorganic chemistry - Ligand field theory and spectroscopy - Biosynthesis of metallo-cofactors - O2 in biology Bioelectrochemistry and Bioenergetics - Electron transfer and redox catalysis - Mitchell hypothesis and theory Enzymology and enzyme kinetics - Enzymes: sources, isolation, assays - Steady-state kinetics - Pre-steady state kinetics - PyMol visualization of 3D enzyme structures - Computational enzymology/ de-novo enzyme design - Membrane enzymology	

LM3741	Fermentation Technology & Environmental Biotechnology	6
Responsible Instructor	Dr. W.M. van Gulik	
Responsible Instructor	Dr.ir. R. Kleerebezem	
Instructor	Dr. W.M. van Gulik	
Instructor	Prof.dr.ir. M.C.M. van Loosdrecht	
Instructor	Dr.ir. R. Kleerebezem	
Instructor	Prof.dr.ir. H.J. Noorman	
Contact Hours / Week x/x/x/x	0/5/2/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Microbial diversity, Black box kinetics of growth/ product formation, mass balances, transport phenomena.	
Course Contents	The FTEB course is divided into two sections: Microorganism characterization (i), and Bioreactor design (ii). Microorganism characterization deals with stoichiometry, kinetics, and thermodynamics of microbial processes in the field of FT and EB. Specific attention is paid to identification of parameters and parameter sensitivity. The bioreactor design section includes batch, fed-batch and continuous bioreactor operation. Specific attention is paid to mixing, gas-liquid mass transfer, biofilm and other biomass retention based bioprocesses, as well as integration of different microorganisms in a single process configuration. All lectures are supplemented with practical examples. Students will elaborate in groups of 3-4 students a case study that should result eventually in the design of a bioprocess. As in the lecture series, students will first elaborate the properties of the microorganisms required and subsequently on the implementation in the bioreactor. Both parts of the case study will be discussed in classical sessions.	
Study Goals	After this course the students are able to analyse and design fermentation and environmental bioprocesses. Most relevant transferable skills: - Problem solving - Teamwork - Communication - Responsibility - Time management - Presentation skills	
Education Method	Lectures and assignments	
Literature and Study Materials	Lecture notes, lecture slides and relevant scientific papers, which will be provided via Brightspace.	
Assessment	Case study (group work) and exam (individual). The scores for the case study and the exam determine for 50% the final grade. The minimum exam score to pass is 5.	
Enrolment / Application	On Brightspace. Not later than one week before the first lecture	
Schedule	Part 1: Microorganism - Introduction - Stoichiometry - Thermodynamics - Basic kinetics - Including product formation - Determining parameters from experimental data - Fed-batch, batch, chemostat - Case study part 1: microorganisms Part 2: Bioreactor - Mass transfer modelling - Design transfer processes in bioreactors - Biofilms structure,function, flux analysis, and process design - Integrating mass transfer, biofilms, and biomass retention - FT-examples - EB-examples - Case study part 2: Bioreactor	
Location	Delft	

LM3751	Transport & Separation	6
Responsible Instructor	Prof.dr.ir. M. Ottens	
Responsible Instructor	Dr. M.E. Klijn	
Module Manager	Dr. M.E. Klijn	
Module Manager	Prof.dr.ir. M. Ottens	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	1. Introductory refreshment of transport phenomena, thermodynamics, and basic concepts of separation processes. 2. Process equipment sizing and link to thermodynamic cycles and energetic yield. 3. Multi-component separations: equilibria, heat and mass transfer in multi-component systems.	
Study Goals	Knowledge, theory and computational skills. At the end of this course, you will be able to LO1. Apply basic principle underlying separation processes for the design of separation equipment LO2. Develop Python-based models for separation equipment used in bioprocessing for simulation purposes LO3. Assess the performance of bioprocess separation steps including mass and energy balances on the basis of its efficiency LO4. Compare separation processes based on their driving force mechanism and calculated (energy) efficiency Transferable skills At the end of this course, you will be able to LO5. Breakdown and discuss computational approaches and theory used to solve a homework assignment via a PowerPoint presentation LO6. Combine problem solving skills and teamwork to complete group assignments	
Education Method	Problem-based and peer learning	
Assessment	Final grade is made of contributions from: 40% group assignments (skills: teamwork and problem solving) 10% exercises presentations (skills: communication and presentation) 50% individual written examination in Python (skills: problem solving) A minimum grade of 5 at the individual written examination is needed to pass the course.	

LM3771	Protein Engineering	3
Responsible Instructor	Dr. F. Hollmann	
Instructor	Dr. F. Hollmann	
Contact Hours / Week x/x/x/x	0/2-4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Elective course of the MSc programme LST.	
Expected prior knowledge	Organic Chemistry (LB1045 or 4051OCSTRY) Chemical Biology (LB2621) OR Catalysis (4052KATALY) Advanced Biocatalysis (LM3731) (or equivalent)	
Course Contents	Protein engineering is an inevitable tool for biocatalysis. Frequently, the wild-type enzymes do not meet the requirements of the desired reaction due to limitations in stability, activity or unsatisfactory selectivity. The key steps in protein engineering including (1) generation of diversity and (2) screening of the diversity to select the most suitable variant. Repetition of these steps is generally performed until the desired trait has been obtained. In this course, the basic principles of protein engineering will be discussed: (1) Generation of diversity through random mutagenesis and or gene shuffling and (semi-)rational protein design (based on e.g. crystal structures or homology models) (2) Analysis of the newly generated diversity with various analytical techniques. In particular: a. Spectrophotometric assays (e.g. NAD(P)H-based UV assays, chromogenic probes) b. Chromatographic assays (gas chromatography, high performance liquid chromatography, electrophoresis) c. NMR- and MS-based assays Furthermore, the pros and cons of the different methods will be discussed (throughput vs precision) as well as strategies to design and select a suitable assay (you get what you screen for).	
Study Goals	After attending this course the student will: - know the basics of the most common protein engineering approaches; - know the most important spectrophotometric and chromatographic screening methods; - be able to choose the most promising protein engineering- and screening method; - be able to critically evaluate the quality of a given strategy; - be able to design new screening strategies.	
Education Method	Lectures - group work - assignments	
Literature and Study Materials	Lecture Sheets, specialized books, articles and patents	
Books	suggested reading: U.T. Bornscheuer et al: Engineering the third wave of biocatalysis Nature, 485, 7397, 185-194. M.T. Reetz: Controlling the enantioselectivity of enzymes by directed evolution: Practical and theoretical ramifications Proc. Natl. Acad. Sci. USA, 2004, 101, 5716-5722.	
Assessment	(1) Written exam (digital off-campus (open book) or on-campus, depending on the COVID-19 measures), 75% of the overall grade (2) Group assignment (3-4 students describing a topic of protein engineering), 25% of the overall grade	

ME45025	Introduction to Multiphase Flow	5
Responsible Instructor	Dr.ir. W.P. Breugem	
Instructor	Prof.dr.ir. R.A.W.M. Henkes	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	A multiphase flow is the simultaneous flow of two or more immiscible phases of matter like gasses, liquids and/or solids. Such flows are important in many environmental and industrial processes and equipment. Examples are gas and oil transport, dredging, steel making, refineries and chemical plants, sediment transport in rivers, cloud formation, etc. This course gives a broad introduction to the fundamental fluid mechanics of such flows. Topics include: classification of multiphase flows into dispersed and separated multiphase flows, relevant dimensionless numbers, stability of gas/liquid two-phase flows, spatial/statistical averaging of the Navier-Stokes equations and continuum modelling, flow pattern maps and mechanistic modelling of multiphase pipe flows, physics of particles (droplets, bubbles, solids), dynamics of particle-laden flow, effects of turbulence, experimental techniques and computational methods for multiphase flows. Examples will be given of the use of multiphase flow theory in industrial applications.	
Study Goals	At the end of this course the student is able to: 1.a classify multiphase flows into separated and dispersed multiphase flows. 1.b formulate appropriate interfacial conditions for a fluid/fluid and a fluid/solid interface. 1.c formulate relevant dimensionless numbers for multiphase flows. 1.d explain the concept of dynamic similarity of multiphase flows. 1.e reflect on scaling choices when designing a laboratory experiment or setting up a numerical simulation. 2.a apply a linear stability analysis to two-phase flows such as for instance a liquid jet in a gas (Rayleigh-Plateau instability) and two horizontal fluid layers in relative motion with each other (Kelvin-Helmholtz instability). 2.b give physical mechanisms for the transition of stratified to non-stratified (bubble or slug) multiphase flows in both (tilted) horizontal and vertical pipes. 3.a apply averaging methods (such as statistical, spatial and temporal averaging methods) to the Navier-Stokes equations for describing the macroscopic behavior of multiphase flows. 3.b formulate appropriate closure models needed to solve the averaged Navier-Stokes equations. 4.a classify gas/liquid pipe flows into stratified, slug and bubble flows, dependent on the superficial gas and liquid velocities. 4.b apply simplified 1D (cross-section averaged) models to each of the flow regimes. 4.c formulate appropriate closure models in order to compute the streamwise pressure drop and liquid holdup in two-phase (gas/liquid) pipe flow. 5.a classify dispersed multiphase flows into flows with solid particles, gas bubbles and/or liquid droplets. 5.b describe similarities and differences between the dynamics of solid particles, gas bubbles and liquid droplets. 5.c apply the Newton-Euler equations of motion to an analysis of the dynamics of solid particles. 5.d describe the settling of a solid sphere under gravity in a stagnant fluid in terms of the Archimedes number and the fluid/solid mass density ratio. 5.e describe possible interactions between solid particles and turbulence of the carrier phase. 6.a explain changes in the shape of a rising gas bubble in a stagnant fluid dependent on the Reynolds number, Eotvos number and Haberman-Morton number. 6.b estimate the terminal rise velocity of a bubble in a stagnant liquid from the equation of motion and an appropriate model for the drag. 6.c give physical mechanisms for the breakup of bubbles/droplets in turbulence. 7. describe the dynamics of some particle-laden flows such as sediment transport through a pipe and/or sediment transport in rivers. 8. describe the merits and shortcomings of different experimental methods for multiphase flows. 9. describe the merits and shortcomings of different numerical methods for simulating multiphase flows. 10.a give examples of multiphase flow in industrial processes. 10.b describe the types of models used to design multiphase industrial processes.	
Education Method	Oral lecture 0/0/2/2; Plenary discussion of homework assignments 0/0/2/2	
Assessment	Written exam. The student can earn bonus points by making homework assignments. The final grade is the sum of the grade for the written exam and the bonus points from the homework.	
Department	3mE Department Process & Energy	

ME45043	Advanced Fluid Dynamics for AP	6
Responsible Instructor	Dr. D.S.W. Tam	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Undergraduate level course in thermo-fluids (WB2542) Knowledge of vector analysis and multivariable calculus is essential for this course (WBMT1050, WBMT2048)	
Course Contents	This course surveys the principal concepts and methods of fluid mechanics. Topics include control volume analysis for mass and momentum, the derivation of Navier-Stokes equations from mass and momentum conservation for incompressible flows, boundary conditions at fluid/fluid and fluid/solid interfaces, dimensional analysis, interfacial tension, inviscid flows, circulation and vorticity, lift and drag, boundary layers, lubrication theory, classical instabilities and waves. Classical solutions of the Navier -Stokes equations are derived and Concepts are illustrated through practical examples from engineering, geophysics and biological fluid dynamics.	
Study Goals	At the end of this class, the student is able to formulate and to use the fundamental governing equations of incompressible, laminar fluid mechanics in classical flow regimes, and to apply these to a new problem in flow technology. More specifically, the student must know how to: 1. formulate the conservation equations for mass and momentum 2. derive the governing equations of motion for an incompressible flow. (Starting from the conservation equations for mass and momentum, the constitutive equation for a Newtonian fluid, and the appropriate boundary conditions at a fluid/solid and fluid/fluid interface) 3. compute the stress tensor and the aero/hydrodynamic forces on a mechanical system 4. derive the vorticity equation and the consequence on the generation and transport of vorticity in 2D (Kelvin circulation theorem) 5. perform a dimensional or scaling analysis to identify the dominant terms in the governing equations. Also, use the non-dimensional numbers introduced in class to determine the flow regime (Reynolds, Strouhal, Froude, Bond, Capillary, Weber) 6. simplify the Navier-Stokes equations for the case of an inviscid flow (the Euler equations), a boundary-layer, a unidirectional flow, a creeping (Stokes) flow and in the lubrication limit. 7. apply appropriately the different forms of the Bernoulli equation The student must be able to derive classical solutions to the Navier-Stokes equations discussed in class. The student must be able to determine whether these classical solutions can be applied to a new problem and use them to compute the flow fields. These solutions include: 8. For ideal flows: a source, a sink, an irrotational vortex, a dipole, corner flows and the superposition of any combination of these using complex potentials (e.g. the 2D irrotational flow around a cylinder and the interpretation of Lift (Joukowski theorem) and Drag (d'Alembert Paradox)) 9. 3D axisymmetric irrotational flow around a sphere for steady and unsteady flows (added mass) 10. For viscous flows: Unidirectional laminar flows, Couette and Poiseuille flows, flow in a lubrication layer 11. Blasius solution for the boundary layer over a flat plate 12. Waves and instabilities in fluid mechanics, gravity waves, Kelvin-Helmholtz instability When facing a new problem in fluid dynamics, the student should be able to: 13. Write all the relevant boundary conditions 14. Use the appropriate formulation of mass and momentum transport (at the control volume level or at the differential level i.e. Navier Stokes equations) 15. Simplify the governing equations using a dimensional/scaling analysis 16. Solve the governing equations for simple flows, when analytically possible 17. Use well-known solutions derived in class to model flows appropriately for the new problem. 18. Model the effect of aero/hydrodynamic forces on the dynamics of a simple mechanical system	
Education Method	Lectures: 3 hours/week Recitations: 1 hour/week Office Hours: 1 hour/week	
Books	Fluid Mechanics Sixth Edition Kundu, Cohen and Dowling	
Assessment	Midterm written Exam (35%) Final written Exam (65%) Semester project (Pass/Fail)	
Remarks	This course is intended for AP master students and follows the schedule of ME45042.	
Department	3mE Department Process & Energy	

ME46000	Nonlinear Mechanics	4
Responsible Instructor	Dr. C. Ayas	
Instructor	Dr. A.M. Aragon	
Instructor	Dr. F. Alijani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	

Course Contents

Introduction

Nonlinear mechanics (NM) builds upon the engineering mechanics (statics and dynamics) courses as embedded within the BSc curriculum. This compulsory MSc course will complete the education in engineering mechanics for a large group of students. Thus, this course should prepare them as good as possible for common problems faced by mechanical engineers without a specialisation in engineering mechanics. On the other hand, there is also a group of students who will continue their training in dynamics and/or statics. For this reason, this course serves as a first step into the more advanced engineering mechanics topics.

Given the broad background of the students who will participate in this course, abstract formulations will be avoided. Moreover, many of the topics will be presented as introductions, creating awareness among the students.

Detailed content:

Introduction to nonlinearities in statics and dynamics
Lagrangian and Eulerian descriptions
Green-Lagrange strain tensor
2nd Piola-Kirchhoff stress tensor and equilibrium equations
Alternative stress and strain tensors,
Geometrical nonlinearity (finite rotations, finite deformations, geometric stiffness effects)

Introduction to nonlinear discrete problems in statics
Computational tools needed to solve nonlinear equations, including topics such as rate equations, physical stiffness, geometric stiffness, iterative solutions procedures, incremental procedures, incremental iterative procedures, path following, dynamic and static transient simulations, convergence, stability points,
Linearized buckling analysis,
Buckling and post-buckling behaviour
Analytical approximations

Introduction to nonlinear material models
Nonlinear elastic models
Simple plasticity models (elastic-ideally plastic, isotropic and kinematic hardening)
Residual stresses and settling
Collapse theorems
Application to simple truss and beam problems
3D plasticity models work (yield surface, flow rule)
Introduction to fracture mechanics

Introduction to nonlinear dynamics
Studying simple discrete nonlinear systems using the phase plane
Quantitative analysis of weakly nonlinear single-degree-of freedom systems using general perturbation theory
Periodic behaviour of simple nonlinear oscillators, limit cycles and softening/hardening responses
Basics of bifurcation theory and stability for simple nonlinear dynamic systems

Study Goals

Learning goals:

Regarding nonlinear mechanics in general, students:
are aware of the range of nonlinearities that are encountered in practical problems, including both static as well as dynamic settings,
are capable of detecting when nonlinear analysis might be required for practical problems,
have awareness of the role of the nonlinearities on both static as well as dynamic systems,
are capable of formulating and solving simple nonlinear engineering mechanics problems.

Regarding geometric nonlinearity (25%), students
can distinguish when geometrical nonlinearity may be relevant (finite rotations, finite deformations, geometric stiffness effects),
can identify effects caused by geometric nonlinearity,
can work with Lagrangian descriptions, are aware of Eulerian descriptions
can evaluate strains (Green-Lagrange),
can evaluate 2nd Piola-Kirchhoff stress tensor and formulate equilibrium equations in nonlinear settings
are aware of alternative stress and strain tensors,
are aware of buckling phenomena and can determine buckling load and post-buckling behaviour for very simple problems,
can make simple estimates, e.g., based on potential energy.

Regarding computational nonlinear mechanics (25%), students
can perform relatively simple nonlinear simulations,
are aware of computational tools needed to solve nonlinear equations, including topics such as rate equations, physical stiffness, geometric stiffness, iterative solutions procedures, incremental procedures, incremental iterative procedures, path following, dynamic and static transient simulations, convergence, stability points,
can identify when buckling might be one of the relevant failure mechanisms and have the knowledge to carry out a linearized buckling analysis,
can make analytical estimates for nonlinear problems, e.g., based on minimum of potential energy.

Regarding physical nonlinearity (25%), students
have awareness of different nonlinear material models,
are familiar and know when to use nonlinear elastic models,
are able to describe the essential features of a nonlinear elastic model,
are aware of simple plasticity models (elastic-ideally plastic, isotropic and kinematic hardening),
are aware and can describe the practical relevance of yielding and the generation of residual stresses and settling,
can make estimates using collapse theorems,
are able to carry out stress analysis for simple problems, e.g., trusses and beams, using simple plasticity models,

	are able to the describe how 3D plasticity models work (yield surface, flow rule), are aware of fracture and can work with linear fracture models. Finally, regarding nonlinear dynamics (25%), students can perform qualitative analysis of simple discrete nonlinear systems using the phase plane, can carry out quantitative analysis of weakly nonlinear single-degree-of freedom systems using general perturbation theory, can discuss the periodic behaviour of simple nonlinear oscillators, limit cycles and softening/hardening responses, are aware of the basics of bifurcation theory and stability for simple nonlinear dynamic systems.
Education Method	Contact hours: Integrated plenary sessions, including lecture and exercises (4 hours per week) Weekly tutor session Study materials: Reader (available in pdf format). lecture slides Suggested literature: Nonlinear Finite Element Analysis of Solids and Structures, 2nd Edition; Rene De Borst, Mike A. Crisfield, Joris J. C. Remmers, Clemens V. Verhoosel; ISBN: 978-0-470-66644-9; 540 pages August 2012 Nonlinear Finite Elements for Continua and Structures, 2nd Edition; Ted Belytschko, Wing Kam Liu, Brian Moran, Khalil Elkhodary; ISBN: 978-1-118-63270-3; 830 pages December 2013, ©2013 Nonlinear Solid Mechanics: A Continuum Approach for Engineering Gerhard A. Holzapfel; ISBN: 978-0-471-82319-3; 470 pages February 2000 Applied Mechanics of Solids. Allan F. Bower, CRC Press, ISBN:978-1-4398-0247-2 (available freely from www.solidmechanics.org) Lecture Notes on Nonlinear Vibrations, Richard H. Rand, freely available at http://audiophile.tam.cornell.edu/randdocs/nlvibe52.pdf. Nayfeh, A. H., & Mook, D. T. (2008). Nonlinear oscillations. John Wiley & Sons.
Assessment	Written (closed-book) exam.
Department	3mE Department Precision & Microsystems Engineering

ME46072	Nonlinear Dynamics	4
Responsible Instructor	Dr. F. Alijani	
Instructor	Dr. G.J. Verbiest	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge of linear dynamics, basics of nonlinear mechanics and differential equations. The essential prerequisite is single-variable calculus, including curve-sketching, Taylor series, and separable differential equations. In a few places, multivariable calculus (partial derivatives, Jacobian matrix, divergence theorem) and linear algebra (eigenvalues and eigenvectors) are used. Fourier analysis is not assumed, and is developed where needed. Introductory physics is used throughout. Other scientific prerequisites would depend on the applications considered, but in all cases, a first course should be adequate preparation.	
Course Contents	Every mechanical and physical system is nonlinear. A thorough understanding of the effects of nonlinearities on the dynamics of these systems is therefore often essential, especially when amplitude of speeds are large. Nonlinear dynamics focuses on the measurement and mathematical description of the motion in nonlinear systems. It thus enables the design of new nonlinear functionalities and maximization of operating ranges while avoiding failure. Nonlinearities cause a wide variety of interesting new physical and dynamic phenomena that cannot be seen in linear systems. Examples include resonances with multiple coexisting solutions and chaotic dynamics. The nonlinear differential equations that govern nonlinear systems have implications that reach far beyond the realm of mechanics, including physics, meteorology, fluid dynamics, chemistry and biology. This course focuses on teaching the fundamental principles of nonlinear dynamics, to bring students to a level where they master the essential concepts of nonlinear dynamics and can apply them to solve practical cases. The course will discuss the following topics: Introduction to nonlinear dynamics together with a brief review of linear vibrations Local geometric theory, stability of equilibrium points and phase space diagrams Limit cycles and self-sustained oscillations Static and dynamic bifurcations Analytic techniques for nonlinear dynamics and perturbation methods Nonlinear resonances: Primary, sub harmonic and super harmonic response, jump phenomena, hysteresis, internal resonance, parametric resonance Computational methods for nonlinear dynamics: Pseudo arc length continuation and shooting method Physical sources of nonlinearities Applications of nonlinear dynamics at the macroscopic and microscopic scale (MEMS/NEMS) Brief introduction to chaos	
Study Goals	In the nonlinear dynamics course, the students will learn to understand, interpret and analyse nonlinear dynamic response of conservative and dissipative mechanical systems in the presence of external excitations. More specifically the students will learn to: 1. Derive nonlinear equations of motion by identifying physical sources of nonlinear dynamics 2. Analyze a diverse variety of dynamic phenomena in nonlinear systems 3. Perform practical and quantitative analyses of nonlinear dynamics systems	
Education Method	Lecture	
Computer Use	The assignment will require using software like Matlab	
Literature and Study Materials	Lecture slides will be made available via Brightspaces	
Books	Nonlinear Dynamics and Chaos With Applications to Physics, Biology, Chemistry, and Engineering Steven H. Strogatz Second Edition July 29, 2014 513 pages https://westviewpress.com/books/nonlinear-dynamics-and-chaos/	
Assessment	Written exam+ assignment.	
Remarks	The course will be taught by Farbod Alijani, and Gerard Verbiest	
Elective	Yes	
Tags	Mathematics Mechanics Physics	
Department	3mE Department Precision & Microsystems Engineering	
Contact	f.alijani@tudelft.nl g.j.verbiest@tudelft.nl	

SC42030	Control for High Resolution Imaging	3
Responsible Instructor	C.S. Smith	
Responsible Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	sc42065	
Expected prior knowledge	Basic knowledge of physics and control theory is expected. The first lectures recap in a condensed form the necessary knowledge in optics. Understanding of signal processing (in particular, the Fourier analysis) is expected for successful work on the homework assignments. The course involves some basic numeric simulations, so practical knowledge of Matlab/python/... is required.	
Course Contents	High resolution imaging is crucial in scientific breakthroughs, such as discovering exoplanets in the habitable zone, or understanding the origin and progress of diseases at a molecular level. For that purpose special optical instruments like Extreme Large Telescopes or STED microscopy are developed. Optical aberrations are the key obstacle that hampers a clear vision and invites control engineers to step in. These are the disturbances induced by the medium, like turbulence in case of astronomy, or by the specimen under investigation, like the change in refraction index due to inhomogeneities in the biological tissue. Adaptive Optics is a modern way to remove the effect of aberrations. This fascinating and expanding field in science is providing an excellent challenge to control engineers to improve the image quality significantly by active control. This course will review the hardware necessary to control light waves in contemporary optical instruments, their modelling from a control engineering perspective, and discuss model-based control methodologies to do disturbance rejection.	
Study Goals	Understand the propagation of light, imaging and aberrations in the imaging process. Understand the operation principle of pupil -plane and focal-plane sensors to estimate the wavefront aberrations. Understand the design principles of opto-mechatronic wavefront corrector devices to correct the wavefront aberrations. Develop spatial and temporal models of complete imaging systems and use these models in the design of model based controllers for aberration correction.	
Education Method	Oral Presentations (lectures), homework assignments (4 in total), self-study in a form of reading and presenting a scientific paper. Two or three seminars (interactive discussion of the homework assignments) are interleaved with the lectures, using the same time slots. Homeworks are allowed to be done in small groups; paper presentations are done in groups.	
Literature and Study Materials	Course Notes	
Assessment	SC42030: (homework and paper presentation) 80% + oral examination 20%.	
Remarks	Old course code: SC4045	
Elective	Yes	
Tags	Broad Mathematics Optimalisation Physics Signals and Systems	
Department	3mE Department Delft Center for Systems and Control	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are expected to submit weekly writing assignments on a subject that is closely related to their field of study. An 85% attendance record is required. This means you may miss not more than one session (preferably none). If you fall behind with your coursework or cannot attend a minimum of six classes, you may be asked to leave the course. You will receive the course credits when all conditions are met.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

WI4011-17	Computational Fluid Dynamics	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor calculus * linear algebra * basic concepts of the finite element method * basic concepts of iterative solvers and preconditioning The student is also expected to be familiar with programming (e.g., in MATLAB or Python).	
Course Contents	Part I: Numerical methods for the convection-diffusion equation; Stability, consistency and convergence of the numerical methods; Extensions to incompressible Navier-Stokes equations; Part II: Basic iterative solvers and preconditioners for the Poisson problem Schwarz domain decomposition preconditioners for the Poisson problem Extension of Schwarz preconditioners to saddle point problems (Stokes and Navier-Stokes equations)	
Study Goals	Learning objectives of the course: Upon completion, the student should be able to: Part I: 1. Derive the equations of fluid dynamics using multivariable calculus and physical conservation laws 2. Explain the singularly-perturbed behavior of solutions to the equations of fluid dynamics 3. Discretize the convection-diffusion equation and derive discretization errors 4. Choose discretizations suitable for convection-dominated flows 5. Discretize the incompressible Navier-Stokes equations Part II: 6. Construct and implement Schwarz domain decomposition methods 7. Explain the concept of the abstract theory of Schwarz methods 8. Apply the abstract Schwarz theory to one- and two-level Schwarz preconditioners to derive condition number bounds for the Poisson problem 9. Apply nonlinear and linear solution techniques to (Navier-)Stokes equations 10. Extend Schwarz methods to (Navier-)Stokes equations	
Education Method	The course comprises 14 lectures (7 lectures for Part I taught by S. Jain, 7 lectures for Part II taught by A. Heinlein).	
Course Relations	In the past WI4011 briefly covered a number of subjects more extensively discussed in the course WI4201 'Scientific Computing'. This overlap has been removed, and therefore student are strongly recommended to take the course WI4201.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * "The Finite Element Method" by TJR Hughes * "Domain Decomposition Methods - Algorithms and Theory" by A. Toselli and O. B. Widlund * "Domain Decomposition Methods for Partial Differential Equations" by A. Valli and A. Quarteroni Additional current topics, which are not covered in standard text books yet, will also be discussed. Further references will be given during the course.	
Assessment	Course will be assessed based on individual assignments (50% of final grade) and project work (50% of final grade; project report and presentation are graded). To pass the course, the following is mandatory. * The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. * The projects must be completed with grades greater than 5.5. No resit option will be available for the assignments, only for the final project. Disclaimer: Information may change depending on the developments around the coronavirus.	
Remarks	This course focuses on the mathematical background of numerical methods for the incompressible Navier-Stokes equations and convection-diffusion equation. For the non-mathematician, it will show how many aspects of the methods can be mathematically analyzed and motivated, far beyond what is shown in more engineering focused courses. For the mathematician, it will show how his knowledge can be applied in one of the most challenging fields of mathematical/computational modeling.	
Co-Instructor	Dr.ir. S. Jain	

WI4014TU	Numerical Analysis	6
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	Classification of partial differential equations and boundary and initial conditions. Maximum principle, uniqueness and non-negativity of the solution, elements of variational calculus. Finite- Difference, Finite-Volume and Finite-Element discretization methods. Solution methods for time-dependent and nonlinear problems. Direct and iterative linear solvers.	
Study Goals	Students will learn: the principles of finite-difference, finite-volume, and finite-element discretizations; the accuracy, consistency, and convergence of numerical solutions; the basic ideas of numerical linear algebra. They will be able to implement a simple PDE solver and to work with existing sophisticated numerical software.	
Education Method	Lectures (possibly, online), homework assignments, programming assignments, written report.	
Literature and Study Materials	[1] J.van Kan, A.Segal, and F.Vermolen, Numerical methods in scientific computing. VSSD, Delft, 2005, improved 2008, ISBN-13 978-90-71301-50-6 [2] Python programming: - Python Help: https://www.python.org/about/help/ - Numpy for Matlab users: https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html [3] FENICS Project: https://fenicsproject.org/ - Installation: https://fenicsproject.org/download/ - Tutorial: https://fenicsproject.org/tutorial/	
Prerequisites	Calculus of functions of several variables, vector calculus, introductory course on numerical analysis (numerical integration, error analysis), introductory course on partial differential equations, some experience with Python or Matlab.	
Assessment	Theoretical and computational assignments, graded reports.	
Tags	Mathematics Modelling Numeric Methods	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced , analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	Take home exam and oral exam. The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. The take-home exam and the oral exam count both for 50% in the final grading of the course. disclaimer: information may change depending on the developments around the coronavirus.	
Special Information	the study guide information may change depending on the developments around the coronavirus.	

WI4201	Scientific Computing	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A basic knowledge on partial differential equations (PDEs), on numerical methods for solving ODEs/PDEs, and on linear algebra.	
Course Contents	During the course, the important steps towards the solution of real-life applications dealing with partial differential equations will be outlined. Based on a well-known basic partial differential equation, which is representative for different application areas, we treat and discuss direct and iterative solution methods from numerical linear algebra in great detail. The discretization of the equation will result in a large system of discrete equations, which can be represented by a sparse matrix. After a discussion of direct solution methods, the iterative solution of such systems of equations is an important step during numerical simulation. Emphasis is laid upon the so-called Krylov subspace methods, like the Conjugate Gradient Methods.	
Study Goals	A. General Study Objectives 1. After having completed this course, the student is able to use finite differences to approximate the solution of a partial differential equation and use an efficient and robust method to compute or approximate the solution of the resulting large system of linear equations. 2. The student will be able to assess the reliability in terms of the accuracy and stability of the numerical approximation of the solution. 3. The student will be able to use methods to approximate eigenvalues of large matrices. B. Specific Study Objectives 1. Finite Difference Method: Boundary Conditions, Grid, Numbering of the unknowns, Error Analysis - The student will be able to discretize 2-dimensional Poisson type problems. The student is able to implement the boundaries in various ways. - The student will be able to analyze the properties of the resulting matrix, as there are symmetry, positive definiteness, M-matrix etc. Also the sparseness of the matrix can be investigated and a notion of band and profile matrices should be known. 2. Direct Solution Methods Gaussian elimination, pivoting, iterative refinement, band and profile methods - The student will be able to understand and use direct solution methods based on Gaussian elimination and LU decomposition. - The student will be able to estimate the error due to rounding errors in the answer and to use pivoting to eliminate most of the errors. - The student will be able to estimate the amount of flops for direct solution methods and how to use more efficient variants as there are Cholesky method, band and profile methods. 3. Basic Iterative Methods - The student will be able to construct Basic Iterative Methods based on the splitting of the matrix. Furthermore an analysis of the convergence could be made. - A comparison of the various method should be possible together with the dependence on the gridsize. - The student will be able to choose suitable starting vectors and stopping criteria. 4. Krylov Subspace methods CG, SPD, preconditioning, general matrices, CGNR, Bi-CGSTAB and GMRES - The student will be able to use and understand the conjugate gradient method, together with a good understanding of the (super)linear convergence. - The student will be able to make a good choice for Krylov Subspace methods in the non-SPD case. - The student will be able to combine Krylov Subspace methods with preconditioners and know three classes of preconditioners: diagonal scaling, BIM and incomplete decompositions. 5. Eigenvalue methods Power method, Lanczos method, Arnoldi method - The student will be able to use the power method and is able to investigate its convergence. - The student will be able to choose suitable starting vectors and stopping criteria. - The student will be able to use the Lanczos and Arnoldi method.	
Education Method	Lectures/computer exercises	
Literature and Study Materials	Lecture notes, for further reading the book Matrix Computations, G.H. Golub and C.F. van Loan, the Johns Hopkins University, Baltimore, 2013, can be used.	
Assessment	The assessment consists of three parts: homework exercises deadline start of Q2 leads to grade G1, take home exam deadline half of January grade G2 and a written exam grade G3. The final grade is (G1+G2+2*G3)/4, provided that all grades are larger than or equal to 5. disclaimer: information may change depending on the developments around the coronavirus.	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4204	Advanced Modeling	6
Responsible Instructor	Dr. J.L.A. Dubbeldam	
Contact Hours / Week x/x/x/x	2e semester intensieve cursus	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The participants will benefit from a solid knowledge of advanced calculus, basic knowledge of ordinary and partial differential equations and programming experience with Matlab (or an equivalent language).	
Course Contents	This modelling course is introducing the principles and practice of mathematical modelling. An important part of the course is to recognize the essential mechanisms governing a phenomenon. These mechanisms have to be translated into mathematical equations and formulas, and included into a mathematical model.This activity requires both a good understanding of the system under consideration and good mathematical skills. Mathematical modelling is essential for understanding processes in real life, complementing expensive experiments that sometimes cannot even be carried out at all. This course will concentrate on applications in physics and life sciences and focus on models in terms of partial differential equations. The course focuses on working in a small group on a specific project. The topics for project work may differ from year to year. For the academic year 2020-2021 some emphasis will be on continuum mechanics	
Study Goals	General 1. Has knowledge of mathematical modelling tools that are used for model applications in physics and for mathematical models formulated in terms of partial differential equations. 2. The student is able to recognize the principles and practice of mathematical modelling. 3. The student is able to identify the essential mechanisms governing a physical phenomenon and is able to translated these into mathematics and can include them into a mathematical model. Specific 1. Is able to make a partial differential equation non-dimensional and can apply scaling techniques to a partial differential equation. 2. Understands conservation principles and constitutive relations. 3. Is able to linearise partial differential equations in order to perform a stability analysis 4. Is able to develop a mathematical model for a practical problem and has the knowledge and maturity to analyse this model with the available mathematical tools. 5. Has the knowledge and skills for the use and assessment of mathematical models and is able to adapt the models for a specific application. 6. Is able to pose adequate questions and has a critical yet constructive attitude towards analysing and solving complex mathematical modelling problems 7. Is able to communicate verbally and in writing about mathematical research with specialists, non-specialists and other parties involved. 8. Has sufficient communicative skills to collaborate with others and discuss mathematical modelling problems and results.	
Education Method	The first lecture summarizes the mathematical modelling tools that might be useful for the projects. Furthermore during this lecture the projects are introduced by the supervisors and the groups of 2-4 students are formed. The main part of the course is offered as a one week "intensive course", fully devoted to the project. In most cases a computer program for simulating a phenomenon from physics of the life sciences has to be developed. At the end of the fourth quarter the written report of the project has to be handed in and the final presentations are scheduled.	
Literature and Study Materials	Many of the students will have a good background in continuum mechanics. The book Continuum Modeling in the Physical Sciences, E. van Groesen and J. Molenaar, Mathematical Modeling and Computation, Vol. 13, SIAM, 2007 (ISBN 978-0-898716-25-2) can be used as background material. Copies of this e-book can be borrowed from the TU-Delft library.	
Assessment	At the first lecture home assignments are given. The final deadline for submitting the home work is the start of the intensive week. The grading will be based on the individual homework (20%) and the project (80%). The grade of the project is based on the written report of the group work and on the oral presentation about the project. The following aspects of the project work will have equal weights: -The development of the model -Mathematical depth -Report and presentation disclaimer: information may change depending on the developments around the coronavirus.	
Pop-up tekst bij intekenen	Welcome to Advanced Modelling!	

WI4205	Applied Finite Elements	6
Responsible Instructor	D. Toshniwal	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The students should have had at least introductory courses on the following topics: * continuum mechanics * partial differential equations * tensor Calculus * linear algebra * numerical methods for ODEs	
Course Contents	Weak formulations (derivation, motivation for use, well-posedness) Abstract introduction to the finite element method (FEM) Theoretical analysis of FEM (e.g., of well-posedness, convergence) Practical implementation of FEM (e.g., for linear/non-linear steady/unsteady problems) in one and higher dimensions Verification and validation of FEM	
Study Goals	After this course, students should be able to: * derive a weak formulation for the give PDE * show the well-posedness of the derived weak formulations * construct isoparametric finite elements in one and higher dimensions * discretize the weak problem using isoparametric finite elements * show the well-posedness of the discretized problem * implement the finite element method * perform basic verification of the implementation correctness	
Education Method	Lectures.	
Literature and Study Materials	The following texts will serve as supplementary references to the class notes: * T.J.R. Hughes. The Finite Element Method: Linear Static and Dynamic Finite Element Analysis, 1987. * P. Wriggers. Nonlinear Finite Element Methods, 2008. Other references will be provided during the course.	
Assessment	Course will be assessed based on assignments (50% of final grade) and a written exam (50% of final grade). Note: Some of the assignments may require the students to study material not covered in class. Similarly, some assignments will ask the students to work on small extensions or generalisations of the concepts covered in class. To pass the course, the following is mandatory: * Assignments: - The assignments (written and programming exercises) must be completed with an average grade greater than 5.5. - The assignments will have deadlines and late submissions will be penalised. - No resit option will be available for the assignments. * Exam: - The written exam must be completed with a grade greater than 5.5. - Resit option will be available for the exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	For more information see the Brightspace course page.	

WI4212	Advanced Numerical Methods	6
Responsible Instructor	Prof.dr.ir. C. Vuik	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Introductory numerical analysis, Introductory partial differential equations, Introductory continuum mechanics.	
Course Contents	This course is an introduction to hyperbolic partial differential equations and a powerful class of numerical methods for approximating their solution, including both linear problems and nonlinear conservation laws. These equations describe a wide range of wave propagation and transport phenomena arising in nearly every scientific and engineering discipline. Several applications are described in a self-contained manner, along with much of the mathematical theory of hyperbolic problems. High -resolution versions of Godunov's method are developed, in which Riemann problems are solved to determine the local wave structure and limiters are then applied to eliminate numerical oscillations. These methods were originally designed to capture shock waves accurately, but are also useful tools for studying linear wave-propagation problems, particularly in heterogeneous material.	
Study Goals	apply numerical methods to hyperbolic systems, study convergence, stability, monotonicity, know methods for 2 dimensional hyperbolic systems	
Education Method	Lectures	
Literature and Study Materials	Finite volume methods for hyperbolic problems R.J. LeVeque Cambridge, UK: Cambridge University Press, 2002. # ISBN-10: 0521009243 # ISBN-13: 978-0521009249	
Assessment	In the first period a set of exercises is given. These should be worked out, strict deadline 26-3-2019 (Grade: G_1). The exercise should be handed in individually. In the middle of April a take home exam including practical exercises is given. This exam can be done by groups of two students. The report of this exam should be returned to us before 11-6-2019 (Grade: G_2). Thereafter an appointment can be made for an oral examination over Chapter 18 (including exercise 18.1, 18.2, and 18.3), 19, and 20 (Grade: G_3). Note that material discussed in Chapter 1 to 9, which is relevant for the material in Chapter 18 to 20 can also be part of the oral examination.The oral exam must be passed with a 6 or higher. Then, provided that G_3 is larger than or equal to 6, the final grade is computed by the formula: (G_1 + G_2 + 2 G_3)/4. disclaimer: information may change depending on the developments around the coronavirus.	

WI4430	Martingales, Brownian Motion	6
Responsible Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	1. Conditional expectation. 2. Martingales, stopping times, optional sampling. 3. Martingale convergence theorems, inequalities. 4. Elementary continuous-time stochastic processes: Poisson process, continuous-time Markov chains. 5. Brownian motion: definition, construction, basic properties. Martingales associated to Brownian motion, brownion motion with drift. 6. Elementary introduction to the stochastic integral, Ito isometry	
Study Goals	1. Understanding the concept of conditional expectation. Being able to compute conditional expectations in concrete examples, and being apply the tower property, and the take-out-what-is-known property. 2. Being able to recognize whether a process is a martingale, sub- or super martingale. 3. Being able to use martingales in computations such as gambler's ruin and distributions of hitting times. 4. Being able to apply the martingale convergence theorems and martingale inequalities in concrete contexts such as law of large numbers, branching process. 5. Being able to perform concrete computations with Brownian motion and its martingales (such as computation of hitting time distribution, recurrence transience). 6. Understanding the definition and construction of the stochastic integral and being able to perform elementary computations with it. 7. Being able to reproduce correctly definitions, statements and proofs and the main definitions and theorems as indicated in a list (this concerns the so-called theory question).	
Education Method	Lectures and problem sessions.	
Literature and Study Materials	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability" for the martingale part of the course. 2. Lecture notes for Brownian motion part of the course. The lecture notes are provided via Brightspace (course content).	
Reader	1. Lecture notes based on chapter 10 of A. Gut ``A graduate course in probability". 2. For the material on Brownian motion we will also sometimes use P. Morters and Y. Peres, Brownian Motion, (2010) freely available at http://people.bath.ac.uk/maspm/book.pdf. Also lecture notes will be available for that part. 3. Other recommended literature: Le Gall: Brownian motion, martingales and stochastic calculus, Springer 2016.	
Assessment	Two graded homeworks and a written exam A homework grade h is computed as the average of the scores on the two graded homeworks. Course grade is the final exam grade f or 0.6f+0.4h, whichever is larger, provided f>5. disclaimer: information may change depending on the developments around the coronavirus. The homework grade does not count for the resit.	

WM-ITAV-4010	Scientific Writing	2
Module Manager	L. Meester	
Instructor	A. Glasbergen-Plas	
Instructor	M. Looij	
Instructor	Drs. B.M.D. van der Laaken	
Instructor	Drs. A.E. Kam	
Instructor	M.J.Y. Wackers	
Instructor	Drs. W.J. Blokzijl	
Instructor	Drs. P.C. Post	
Instructor	S. Baars	
Instructor	Drs. M. Bliekendaal	
Instructor	L.C. Schroten	
Co-responsible for assignments	Drs. B.M.D. van der Laaken	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1 2	
Exam Period	none	
Course Language	English	
Course Contents	In this course, you learn to write a scientific article, either a research article based on your own research data or a literature review about a subject of your own choice. This is a necessary skill for anyone who wants to pursue an academic career after their graduation, but it can also be used immediately for all academic texts you will write during your Master program, such as your Master thesis. In seven weeks, we will go through all steps of the writing process, from formulating a good main question and finding relevant literature, to the actual writing, re-writing and final editing. You will exercise with finding, reading and managing relevant academic literature, writing in an academic style, building a comprehensive argumentation, reviewing fellow students' articles and using other students and the instructor's comments to improve your own work.	
Study Goals	The purpose of this course is to learn how to write a scientific text. To achieve this, at the end of this course you will: know what the main characteristics are of a scientific text be able to formulate a main question be able to find, critically read and manage scientific literature be able to use literature properly and avoid plagiarism be able to build up your argumentation be able to structure an article according to the conventions in your field of study be able to use scientific English style be able to use tables and figures to support and communicate your results be able to give feedback on somebody elses article be able to use feedback for improving your work.	
Education Method	(Online and/or on campus) practical, in 6 sessions (attendance mandatory). Every week you have to read some background information about scientific writing and hand in a part of your text. Participants must attend all sessions - one missed session is allowed only - and hand in all assignments in time. Students who receive a pass for this course are rewarded with 2 ects. This equals 56 hours of study. A total of 12 hours is spent on attending (online) classes, in which you can ask questions, discuss the feedback you received on your work and discuss aspects of scientific writing with your fellow students; the remaining 44 hours is for self study, writing and revising. In seven weeks, from preparing lecture 2 up to handing in your final article in week 8, you will have to spend at least 6 hours of self study on this course, every week. It is important that you make sure you have this time in your personal schedule. At the beginning of the course you will mainly be reading up about scientific writing and the subject of your text. As the course proceeds, you will be spending more of your time on writing, giving feedback and revising your own text.	
Books	Theory about academic writing will be made available through Brightspace.	
Assessment	You write a scientific article of 2000-2400 words (excluding the list of references and the abstract). You have to hand in parts of your article every week. Your final grade is based on the final article. An evaluation form for the grading of the article is available throughout the course.	
Elective	Yes	
Category	MSc level	

WM0201TU-ENG	Technical Writing	2
Module Manager	Drs. J.A. Weitenberg	
Instructor	Drs. J.A. Weitenberg	
Co-responsible for assignments	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is geared to two groups of writers: engineers who need to write technical reports and scientists who need to write scientific articles. Modern engineers are required to possess excellent communication skills. Not only do they need to know their business, but they also need to know how to communicate this business to other experts, decision makers, and implementers. They need to be able to write accessible technical reports and proposals and to hit the correct tone in professional correspondence. The goal of scientific research is publication. Scientists are rarely measured by their knowledge of science; instead, they are measured and become known (or remain unknown) by their publications. Scientific articles need to be well structured to allow for efficient reading and accessible to the scientific community. They should also be written in a professional style, convincing, effective, and without language errors. Both groups will benefit from this course, which focuses on both the quality of the text and on the writing process itself. The following subjects will be dealt with: a systematic approach to writing requirements to technical reports and scientific papers and their components structuring texts on several levels writing convincingly writing "proper and appropriate" language referencing Seven lectures will be used for instruction, feedback, and practice. Students will be required to hand in several assignments while they are working on the technical or scientific report that needs to be written to complete the course. No more than one lecture may be missed.	
Study Goals	Improving written communication skills.	
Education Method	Workshop. Writing assignments, peer feedback.	
Literature and Study Materials	Elling et al (2011): Report Writing for Readers with Little Time. 1st edition, Noordhof Uitgevers. All other materials are available in Brightspace.	
Assessment	Several smaller assignments and one final report or scientific paper.	
Enrolment / Application	Brightspace	
Elective	Yes	
Category	MSc level	

WM0320TU	Ethics and Engineering	3
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course is identical to the initial part of the course WM0329TU. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%), presentation and active participation during the working group sessions (40%). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enrol as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course wm0329tu (6 ects).	
Elective	Yes	
Category	MSc niveau	

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Master Nanobiology**

2nd year Msc NB 2022	
Responsible Program Employee	J.C. Colgrove
Director of Studies	T. Idema
Program Coordinator	J.C. Colgrove
Program Coordinator	D. Mevius
Contact for students	info-msc-nb@tudelft.nl
Introduction 1	Students who began the MSc Nanobiology prior to 2022-2023 will have a different structure for their programme. Please see the programme specific appendix for details of the courses to follow. And contact the programme coordinator if you have any questions.

NB5015	Seminars	2
Responsible Instructor	I. Chaves	
Contact Hours / Week x/x/x/x	1/1/1/1	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	At the beginning of this course in Q1 there will be 3 in person sessions. Attendance is highly encouraged but not mandatory. Session 1: Introductions and expectations, how to find seminars, form groups, what should be included in a report. Session 2: A faculty seminar on a nanobiology topic Session 3: Share and discuss the seminar reports from session 2. Attend seminars and write seminar reports on your own schedule for the rest.	
Study Goals	To develop skills in gathering information and knowledge from the attendance of scientific seminars.	
Education Method	Extracting information from seminars and reporting on them in writing. You will write an one-page report (~500 words) on each of the 12 attended seminars. Illustrations may be included.	
Assessment	A faculty member will read your reports (a grade will be uploaded within 2 weeks of receiving the last report). This is a pass/fail course. The report should summarize the contents of the seminar, and include (at the end) a personal reflection (discuss whether it was interesting, useful, applicable to your current research, whether you could understand what was presented). The reports will be checked for plagiarism. Reports are submitted via Brightspace.	
Special Information	The teacher can be reached at i.chaves@erasmusmc.nl	
Location	Intro sessions are at Erasmus MC.	

NB5020	Literature Review Report	4
Responsible Instructor	Dr. R.J.T.S. Palstra	
Course Coordinator	Dr. R.J.T.S. Palstra	
Contact Hours / Week x/x/x/x	1/1/1/1	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	You will write an insightful essay/short review on a specific topic within the field of Nanobiology. You can contact a member of the Nanobiology faculty on the green list to discuss an appropriate topic and receive literature suggestions on the topic. Your MEP supervisor can supervise this writing assignment and is likely the best choice.	
Study Goals	To learn how to gather literature on a chosen topic, to extract relevant information, and to report your findings in a written mini-review. At the completion of this course the student will be able to: 1. Collect scientific literature on a specific topic and evaluate its usefulness. 2. Extract information relevant to a specific topic from a body of scientific literature. 3. Organize information into a short review understandable to a general science audience 4. Prepare a short review according to a given format, including referencing style and preparation of display items (if used) in the same manner expected for publishing such a review.	
Education Method	On-line resources provided for self study. Consultation with teachers for advise. You will be guided and receive feed-back on the written report by the Nanobiology faculty member whose topic you chose.	
Assessment	The written essay/review will be assessed and graded (1-10) by a faculty member assigned by the course coordinator. It is expected that the written assignment will be completed by June of the year in which this course taken, usually the second year of the MSc program.	
Special Information	This course will be offered for the last time in 2022-2023. Students who complete it should also complete NB5030 Project Proposal Writing and NB5040 Research Presentation and take NB5900 Master End Project (44EC). Students starting the MSc in 2022-2023 or starting their MEP in Q3 of 20222-2023 or later will take the new courses NB4510 Project Development 1, NB4520 Project Development 2 and NB5942 Master End Project (42 EC) unless they receive special permission from the Nanobiology Academic Counselor. The teacher can be contacted at r.palstra@erasmusmc.nl	

NB5030	Project Proposal Writing	2
Responsible Instructor	Dr.ir. W.M. Baarends	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course is connected to your Master End Project. please note that you need a Master End Project subject before you can start this course (It is not necessary that you already started your MEP project). This course will be taught for the last time in Q1-Q2 (September -December) of 2022-2023. In regular course meetings you are educated on how to write a scientific proposal. You will develop a first draft of your own research proposal, describing the plan of work for the one-year period leading to the MSc thesis. The written research proposal will be prepared together with, and approved by, your research supervisor. This plan, agreed upon by both student and faculty, will be the basis for your research work leading to the MSc Nanobiology degree. The research proposal should describe the plan of work for a full period of research. This will include: Title, Summary, Specific Aims, Introduction, Experimental Plan, and Reference list (including figures as appropriate). These parts will be written in a step -wise manner. At each stage, the written parts will be evaluated by faculty on specific elements of scientific writing, as well as content, and revised accordingly. Students will also review and evaluate the completed proposals of two of their peers to provide comments, during a discussion session, before the final revision. The final proposal follows a template with word count limitations and will probably be in the range of 6-10 pages long. Deadlines for submitting the course proposal are in Brightspace. Please enroll in the course as soon as possible, but at the latest 2 weeks before the start of the course, and complete the assignment: MEP project title+supervisor+location (title can be a draft title).	
Study Goals	Upon completion of this course you will be able to: Find relevant scientific literature and summarize it in comprehensible and structured written scientific English to provide background information on the topic to be studied in your Nanobiology Master research project (NB5900) To clearly and convincingly state the main scientific question, and specific objectives, that will be addressed To identify and advertise the innovative aspects of the research project in written form Describe a realistic experimental approach and time plan to answer the research question. Identify and convincingly describe the possible knowledge utilisation aspects of the expected results of the research project Combine all the above products into a complete research project proposal such as those prepared in order to obtain funds from granting agencies. To critically evaluate such a proposal from peers and propose improvement during an evaluation meeting To constructively incorporate feedback to improve the scientific proposal	
Education Method	(possibly partly online) Workgroup sessions, peer review session, and writing assignment	
Literature and Study Materials	The Elements of Style, 4th Ed W. Strunk Jr. and E.B. White, Longman Publishers, 2000	
Books	The Elements of Style, 4th Ed W. Strunk Jr. and E.B. White, Longman Publishers, 2000	
Assessment	The evaluation of the written research proposal assignment will be based on organization, clarity and grammar of the writing,and significance, methodology and feasibility of content and logic, and the use of English. Attention will be paid to research questions posed, the appropriateness of the experimental approach, feasibility of the work for a one-year project, and suitability of the work for completion of the MSc thesis.The writing assignment will be evaluated by faculty at each stage to provide comments for correcting, revising and improving the proposal. The complete draft proposal will be reviewed by 2 other students in the class to provide additional comments for revision. The final proposal is formally graded by one of the faculty, using a rubric (available on brightspace) whereby 8 categories with learning outcomes are addressed for assessment: 1;proposal structure, 2;language/grammar, 3;abstract (1-3 together 30%), 4;title, background, main question, specific objectives (20%), 5;proposed experimental approach and feasibility (20%), 6;innovation and scientific impact (10%), 7;knowledge utilization (10%), 8;providing feedback to peers (5%), 9;handling feedback (5%). You will receive a grade on a scale from 1 (worst) to 10 (best). In order to pass, no more than 1 item of the 4 categories may receive a grade lower than 5.5, and the overall grade must be 5.8 or higher. Re-exam in case of a final grade <5.8 consists of one extra possibility for revision, whereby the deadline for submission will be set in consultation with supervisor and faculty of the course, but no later than one month after the grade was received.	
Special Information	This course will be offered for the last time in fall of 2022-2023. Students who complete it should also complete NB5020 Literature Review Report and NB5040 Research Presentation and take NB5900 Master End Project (44EC). Students starting the MSc in 2022-2023 or starting their MEP in Q3 of 20222-2023 or later will take the new courses NB4510 Project Development 1, NB4520 Project Development 2 and NB5942 Master End Project (42 EC) unless they receive special permission from the Nanobiology Academic Counselor.	

NB5040	Research Presentation	2
Responsible Instructor	S. Rao	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	To advance and broaden skills in oral presentations you will prepare and present a 15-20 minute talk. You can choose to present your research project progress. Alternatively you can choose to try out a different style of presentation such a proposing a new research project, a business plan using your NB skills or others arranged in consultation with the instructor. A research project presentation should include a brief introduction to the project, some information on specific experiments, details on methods, controls, and actual data, and a conclusion with a brief statement on the results obtained. The content and aim of a different style presentation are agreed on ahead of time between the student and instructor. Giving presentations on-line is a newly valuable skill, attention to and opportunity to practice on-line presentations is now added to this course.	
Study Goals	At the end of this course the student will be able to: 1. Organize experimental data, or new ideas, in a concise and informative presentation. 2. Prepare oral script and visual aids for a presentation of a specific time duration. 3. Deliver an oral presentation on a chosen topic appropriate to a specified audience. 4. Effectively communicate experimental results, or logical arguments in an oral presentation. 5. Design a presentation for a specific audience. 6. Provide constructive feedback on presentation content and style to peers	
Education Method	Organizational meeting to plan and schedule presentations. Short lectures on specific aspects of presentation skills from experienced speakers. On line instructions including recorded instructions and relevant resources available via web site links. students prepare a presentation based on self study from provided sources, consultation with faculty and their peers. Each student will prepare and deliver a presentation to a group of peers. Each student will attend all presentations of peers and provide constructive feedback.	
Assessment	You will give one or two 15 to 20 minute talks in the format and style agreed on with the instructor, accompanied by slides or other visual aids. This will be evaluated based on clarity, organization and presentation performance aspects. The oral presentation will be assessed by a faculty members and peer reviewed by fellow MSc students. This course is evaluated pass/fail. To receive credit you must give a presentation and provide peer review of the presentation of all other students. NOTE: Due to current restrictions on in person meetings and instruction this course will be largely organized on-line. It is possible that all presentations will be given on line or that there will be a combination of on-line and live presentations.	
Enrolment / Application	Enrol via Brightspace. Students must enroll before week 1.1 (Q1 course) or 3.1 (Q3 course) in order to be informed in time. Exact details of timing will be announced in the Brightspace page. The course activities consist of an information session and some sessions for students to give presentations and do peer review. Schedule is mutually decided upon by students and instructor.	
Special Information	This course will be offered for the last time in 2022-2023, it will be offered twice, once in Q1 and once in Q3. Students who complete it should also complete NB5030 Project Proposal Writing and NB5020 Literature Review Report and take NB5900 Master End Project (44EC). Students starting the MSc in 2022-2023 or starting their MEP in Q3 of 20222-2023 or later will take the new courses NB4510 Project Development 1, NB4520 Project Development 2 and NB5942 Master End Project (42 EC) unless they receive special permission from the Nanobiology Academic Counselor.	
Location	Erasmus MC. The teacher can be reached at s.rao@erasmusmc.nl	

NB5900	Master End Project Nanobiology	44

Responsible Instructor	T. Idema
Course Coordinator	L. van der Elst
Course Coordinator	J.C. Colgrove
Course Coordinator	Drs. E. Jacobs
Contact Hours / Week x/x/x/x	30/30/30/30
Education Period	None (Self Study)
Start Education	1 2 3 4
Exam Period	Different, to be announced
Course Language	English

Course Contents

NB5900 Master End Project is for students who began the Nanobiology programme in academic year 2021-2022 or earlier. New curriculum beginning in 2022-2023 will lead to a different Master End Project course code (NB5942) with different project development courses.

The Master End Project (MEP) is usually conducted in the second year of the MSc. It will take approximately 10 months, and is conducted in a host research laboratory of your choice from the Nanobiology GreenList. Projects outside the GreenList require Board of Examiner approval. The research project comprises 44 ECTS and includes writing a Masters thesis, presenting the work and defending it orally.

This MEP is connected to the Master course Project Proposal Writing (NB5030). Please note that you have to follow this course at the start of your MEP or from the moment that you have a Master End Project subject (for NB5030 it is not necessary that you already started your MEP). This course will be available in fall of 2022 for the last time.

During your MEP, you work as a member of a research group and collaborate with other students, PhD students, postdocs, technicians and group leaders. You will be working on a specific research topic of your choice, doing experiments, simulations, or theoretical work, and generating new ideas for further research. You are expected to implement the knowledge from the Bachelor and Master phase and apply that knowledge to the research subject. If the research project is outside the Erasmus MC or TU Delft, you need to have both an internal (green list) and external supervisor.

The research project is concluded by submission and defense of the MSc thesis. The thesis will take the form of a full-length research article, suitable for publication in an international journal. In contrast to a regular article manuscript, the MSc thesis will include a more elaborate introduction, describing the scientific background of the study, and the methods section will be expanded to include a complete and detailed description of all methods that have been applied. The MSc thesis should demonstrate the ability of the student to organize and present results and knowledge in a form required for publication of a scientific article.

The defense of the thesis will take about 30 minutes, and will be preceded by a public presentation (of also roughly 30 minutes) on the aim, results and conclusions of the thesis work.

Study Goals

To obtain practical experience in laboratory or theoretical research, to collect scientific data, to practice communication skills, master research techniques, and further develop scientific thinking and reasoning.

Education Method

Lab or theory work at university research groups. An internal university research project is guided by an internal (Nanobiology green list) supervisor and evaluated by a thesis committee. For research projects at external universities and research institutions, or a lab with a PI not on the Nanobiology green list, the supervisor from the host group is responsible for day-to-day supervision. During the research project the student will report to the Erasmus MC / TU Delft supervisor on a regular basis about progress made on the project. In practice, the project may be co-supervised by a PhD student or postdoc from the main supervisor's research group.

Prerequisites

Students may start their Masters End Project if they:

* have been admitted to the Nanobiology Masters programme,

* have passed bridging/homologation requirements or other obligations from the bachelor programme,

* have passed at least 35 EC of Nanobiology MSc courses, not including Internship or Academic Research project. These credits include electives.

* have submitted and had approved the appropriate approval form.

Assessment

The assessment committee will consist of at least three members. At least two members belong to academic staff (tenured or tenure track) not limited to TU Delft, Erasmus MC, nor Leiden University or members for the occasion appointed by the Board of Examiners (art 5.2)3. The third member can be a member of academic staff or a researcher with a PhD employed at TU Delft or Erasmus MC (for Nanobiology). Together they satisfy the criteria a to d. Committee members that do not meet the criteria, including PhD students, can be added as fourth or fifth member of the committee.

a. One of the members is the responsible thesis supervisor and must be on the Nanobiology GreenList. In case of two supervisors both are members of the assessment committee.
b. One of the members belongs to the teaching staff of the MSc programme, as identified on the Nanobiology GreenList.
c. One of the members is a full professor or an associate professor with ius promovendi;
d. One of the members of academic staff belongs to a TU Delft or Erasmus MC research group or research entity independent of the responsible thesis supervisor.
If the composition of the assessment committee does not meet the regulations mentioned, the proposal must be submitted to Board of Examiners at least one month prior to the defense.

Daily supervisors (PhD students, technicians) can be added to the committee, be involved in the defense, and be invited to give their opinion on the performance of the student (and their suggestion for a grade) but (unless explicitly appointed by the Board of Examiners as a member of the committee) have no formal vote on the grade.

During the research project the students will be judged on:
- Performance in regular tasks
- Active participation in the project and research group
- Taking initiative
- Communication within the research group
- Oral presentation and defense
- Written report

The committee will use the MEP grading rubric from the faculty of Applied Sciences to determine the final grade. See the TNW Thesis Brightspace page for the full rubric.

Enrolment / Application	Find a host group that appeals to you and has room for a student project. Together with your intended supervisor, fill out the "Master end project application form. The form must be submitted and accepted prior to the start of your Master end project.
Special Information	A research project can be completed under the supervision of a TU Delft or Erasmus MC PI found on the Nanobiology GreenList. The GreenList which can be found on the Brightspace page of the Thesis office Applied Science. Students may do projects in research gorups outside the GreenList but this requires, a Nanobiology supervisor and approval by the Board of Examiners prior to beginning the project. Concerning Internship, Academic Research Project and Master End Project within the Master Nanobiology: - At least one of these must be carried out in a research group headed by a Nanobiology GreenList supervisor. - The two projects may not be done within the same research group.

NB5942	Master End Project Nanobiology	42
Responsible Instructor	T. Idema	
Course Coordinator	Drs. E. Jacobs	
Course Coordinator	L. van der Elst	
Course Coordinator	J.C. Colgrove	
Contact Hours / Week x/x/x/x	0/40/40/40	
Education Period	None (Self Study)	
Start Education	1 2 3 4	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The Master End Project (MEP) is usually conducted in the second year of the MSc. It will take approximately 8 months, and is conducted in a research group on the Nanobiology GreenList. The research project comprises 42 ECTS and includes writing a Masters thesis, presenting the work and defending it orally. During your MEP, you work as a member of a research group and collaborate with other students, PhD students, postdocs, technicians and group leaders. You will be working on a specific research topic of your choice, doing experiments, simulations, or theoretical work, and generating new ideas for further research. You are expected to implement the knowledge from the Bachelor and Master phase and apply that knowledge to the research subject. If the research project is outside the Nanobiology programme GreenList, you need to have both an internal (GreenList) and external supervisor. Projects outside the GreenList must be approved by the Board of Examiners prior to beginning. The research project is concluded by submission and defense of the MSc thesis. The thesis will take the form of a full-length research article, suitable for publication in an international journal. In contrast to a regular article manuscript, the MSc thesis will include a more elaborate introduction, describing the scientific background of the study, and the methods section will be expanded to include a complete and detailed description of all methods that have been applied. The MSc thesis should demonstrate the ability of the student to organize and present results and knowledge in a form required for publication of a scientific article. The defense of the thesis will take about 30 minutes, and will be preceded by a public presentation (of also roughly 30 minutes) on the aim, results and conclusions of the thesis work.	
Course Contents Continuation	A research project can be followed with a supervisor on the Nanobiology GreenList that can be found on the Brightspace page of the Thesis office Applied Science. With approval from the Board of Examiners, students may apply to do a project outside the Nanobiology programme. When planning research projects, students must do either their MEP or their Broadening Research Project internal to the Nanobiology programme, they can do both. Please see the programme page for Broadening Research Projects to see which allowed projects are considered internal and other rules about the projects.	
Study Goals	To obtain practical experience in laboratory or theoretical research, to collect scientific data, to practice communication skills, master research techniques, and further develop scientific thinking and reasoning.	
Education Method	Lab or theory work at university research groups. An internal university research project is guided by an internal (Nanobiology GreenList) supervisor and evaluated by a thesis committee. For research projects at external universities and research institutions, or a lab with a PI not on the Nanobiology GreenList, the supervisor from the host group is responsible for day-to-day supervision. During the research project the student will report to the Erasmus MC / TU Delft supervisor on a regular basis about progress made on the project. In practice, the project may have a daily supervisor who is a a PhD student or postdoc from the primary supervisor's research group.	
Prerequisites	Students may start their Masters End Project if they: * have been admitted to the Nanobiology Masters programme, * have passed any bridging/homologation requirements or other obligations from the bachelor programme, * have passed at least 35 EC of Nanobiology MSc courses, not including broadening research projects. These credits can include electives. * have passed NB4510 Project Development Part 1 and NB4520 Project Development Part 2 * have submitted and had approved the appropriate approval form.	
Assessment	The assessment committee will consist of at least three members. At least two members belong to academic staff (tenured or tenure track) not limited to TU Delft, Erasmus MC, nor Leiden University or members for the occasion appointed by the Board of Examiners (art 5.2)3. The third member can be a member of academic staff or a researcher with a PhD employed at TU Delft or Erasmus MC (for Nanobiology). Together they satisfy the criteria a to d. Committee members that do not meet the criteria, including PhD students, can be added as fourth or fifth member of the committee. a. One of the members is the responsible thesis supervisor and must be on the Nanobiology GreenList. In case of two supervisors both are members of the assessment committee. b. One of the members belongs to the teaching staff of the MSc programme, as identified on the Nanobiology GreenList. c. One of the members is a full professor or an associate professor with ius promovendi; d. One of the members of academic staff belongs to a TU Delft or Erasmus MC research group or research entity independent of the responsible thesis supervisor. If the composition of the assessment committee does not meet the regulations mentioned, the proposal must be submitted to Board of Examiners at least one month prior to the defense. Daily supervisors (PhD students, technicians) can be added to the committee, be involved in the defense, and be invited to give their opinion on the performance of the student (and their suggestion for a grade) but (unless explicitly appointed by the Board of Examiners as a member of the committee) have no formal vote on the grade. During the research project the students will be judged on: - Performance in regular tasks - Active participation in the project and research group - Taking initiative - Communication within the research group - Oral presentation and defense - Written report The committee will use the MEP grading rubric from the faculty of Applied Sciences to determine the final grade. See the TNW	

	Thesis Brightspace page for the full rubric.
Enrolment / Application	Find a research group that appeals to you and has room for a student project. Together with your intended supervisor, fill out the "Master end project application form. The form must be submitted and accepted prior to the start of your Master end project.
Special Information	Article 24A MSc thesis projects outside the designated research groups* 1. Students who wish to spend more than three months working on the MSc thesis project outside research groups designated in the Nanobiology GreenList must obtain permission from the Nanobiology Sub-Board of Examiners. 2. To obtain the permission mentioned in section 1, students must submit a written request to the Sub-Board of Examiners with the consent of the responsible thesis project supervisor* at least 20 working days before starting the MSc thesis project. 3. This request is subject to the following criteria and should be submitted by the GreenList supervisor to the Board of Examiners: a. A thesis project is carried out under supervision of designated research groups; b. The project is part of existing research; c. The initiative for collaborating with third parties rests with the research group, not with the student; d. The project needs to be of the same academic level as that of a project carried out completely in one of the research groups; e. For a thesis project the faculty provides intensive and academic supervision. This means that there has to be frequent consultation between the supervisor and the student; f. For the assessment, article 24e. applies *designated research groups means groups on the Nanobiology GreenList *Every project must have GreenList responsible supervisor.

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